

Chapter 52

Ethernet Media Access Controller (EMAC)

52.1 Overview

By using the external Ethernet PHY (physical layer), ESP32-P4 can send and receive data via Ethernet MAC (Media Access Controller) according to the IEEE 802.3 standard, as Figure 52.1-1 shows. Ethernet is currently the most commonly used network protocol that controls how data is transmitted over local- and wide-area networks, abbreviated as LAN and WAN, respectively.

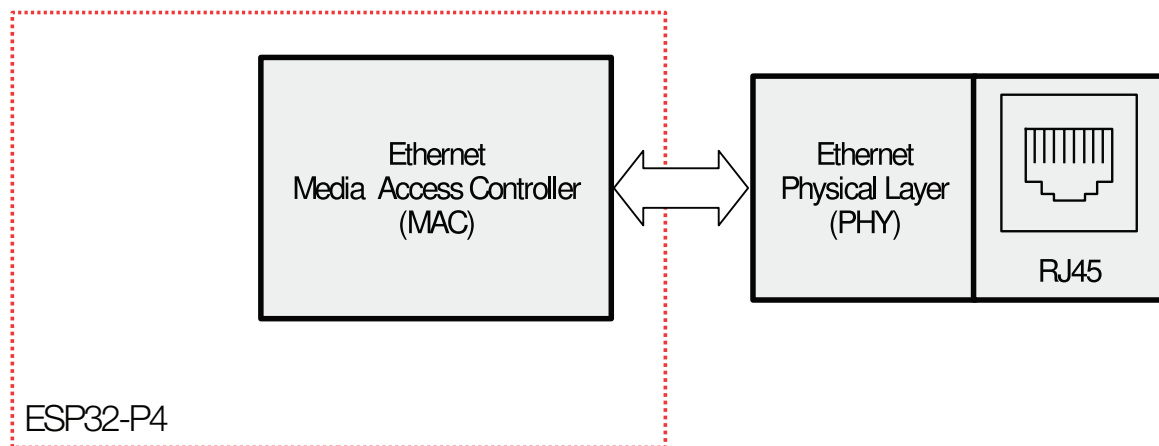


Figure 52.1-1. Ethernet MAC Functionality Overview

ESP32-P4 Ethernet MAC complies with the following standards:

- IEEE 802.3-2002 for Ethernet MAC
- IEEE 1588-2008 standard for precise networked clock synchronization
- IEEE 802.3 standard Media Independent Interface (MII) and Reduced Media Independent Interface (RMII)
- IEEE 802.3az-2010 for Energy Efficient Ethernet
- IEEE 802.1Q for VLAN frame format

52.2 Features

- Data rates of 10/100 Mbit/s through an external PHY interface
- Communication with an external Fast Ethernet PHY through IEEE 802.3-compliant MII and RMII interfaces
- Full-duplex and half-duplex modes

- Carrier Sense Multiple Access or Collision Detection (CSMA/CD) protocol in half-duplex mode
- IEEE 802.3x flow control in full-duplex mode
- Optional forwarding of received pause control frame to the user application in full-duplex mode
- Back-pressure flow control in half-duplex mode
- Automatic transmission of zero-quanta pause frame on deassertion of flow control input in full-duplex mode
- Preamble and start-of-frame data (SFD) insertion in Transmit, and deletion in Receive paths
- Automatic CRC and padding (all 0) generation controllable on a per-frame basis
- Options for automatic padding generation for data below the minimum frame length
- Programmable frame length supporting jumbo frames of up to 16 KB
- Programmable inter-frame gap (IFG) from 40 to 96 bit times in steps of 8)
- Supports a variety of flexible address filtering modes:
 - Up to eight 48-bit perfect address filters with masks for each byte
 - Up to eight 48-bit source address (SA) address comparison check with masks for each byte
 - Option to pass all multicast addressed frames
 - Promiscuous mode support to pass all frames without any filtering for network monitoring
 - Passes all incoming packets (as per filter) with a status report
- Separate 32-bit status returned for transmission and reception packets
- Supports IEEE 802.1Q VLAN tag detection for reception frames
- Separate transmission, reception, and control interfaces for the application
- Management Data Input/Output (MDIO) interface for PHY device configuration and management
- Supports checksum off-load for received IPv4 and TCP packets encapsulated by the Ethernet frame
- Supports checking IPv4 header checksum and TCP, UDP, or ICMP checksum encapsulated in IPv4 or IPv6 datagrams
- Supports 64-bit timestamp on each transmit and receive frame (see IEEE 1588-2008)
- Supports Energy Efficient Ethernet (see IEEE 802.3az-2010)
- CRC replacement, SA insertion or replacement, and VLAN insertion, replacement or deletion in transmit frames
- Two FIFOs: 256-byte TX FIFO and 256-byte RX FIFO
- Receive status vectors inserted into RX FIFO after the EOF (end of frame) transfer, which enables multiple-frame storage in RX FIFO without requiring another FIFO to store those frames' receive status
- Option to forward good runt frames
- Supports statistics by generating pulses for frames dropped or corrupted due to overflow in RX FIFO
- Automatic re-transmission of collision frames (subject to certain conditions, see Section [52.4.1.1](#))

- Discarding frames in cases of late collisions, excessive collisions, excessive deferrals, and underflow conditions
- Software control to flush TX FIFO

52.3 Ethernet MAC Architecture

Figure 52.3-1 shows the block diagram of the Ethernet MAC.

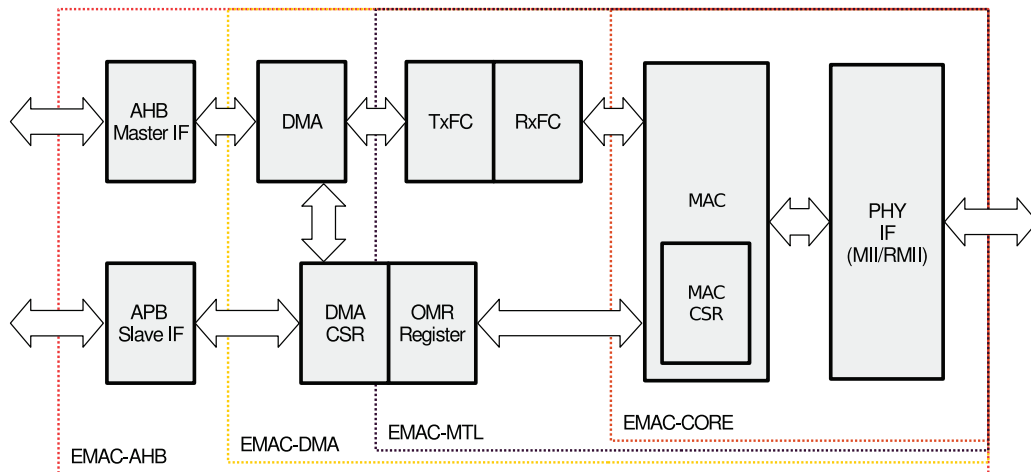


Figure 52.3-1. Ethernet MAC Block Diagram

Ethernet MAC consists of three layers: EMAC_CORE (MAC Core Layer), EMAC_MTL (MAC Transition Layer), and EMAC_DMA (Direct Memory Access). Each of these three layers has two directions: TX and RX. They are connected to the system through the Advanced High Performance Bus (AHB) and the Advanced Peripheral Bus (APB) on the chip. Off the chip, they communicate with the external PHY through the MII and RMII interfaces to establish an Ethernet connection.

52.4 Functional Description

52.4.1 EMAC_CORE

The MAC supports two interfaces (see Section 52.4.4) towards the PHY chip. The PHY interface can be selected only once after the chip reset. The MAC core communicates with the application side (DMA side) using the MAC Transmit Interface (MTI), MAC Receive Interface (MRI), and the MAC Control Interface (MCI).

52.4.1.1 Transmission

Transmission is initiated when the MTL application pushes in data with the SOF signal (start of frame) asserted. When the SOF signal is detected, the MAC accepts the data and begins transmitting to the RMII or MII. The time required to transmit the frame data to the RMII or MII after the application initiates transmission is variable, depending on delay factors like IFG delay, time to transmit preamble or SFD (Start Frame Delimiter), and any back-off delays for half-duplex mode. Until then, the MAC does not accept the data received from MTL by deasserting the ready signal.

After the EOF (end of frame) is transmitted to the MAC, the MAC completes normal transmission and then gives the Transmit Status to the MTL. If a normal collision (in half-duplex mode) occurs during transmission, the MAC core makes valid the Transmit Status to the MTL. It will then accept and drop all further data until the next SOF is received. The MTL block should retransmit the same frame from SOF on observing a retry request (in the Status) from the MAC.

The MAC issues an underflow status if the MTL is not able to provide the data continuously during the transmission. During the normal transmission of a frame from the MTL, if the MAC receives an SOF without getting an EOF for the previous frame, then the SOF is ignored and the new frame is considered as a continuation of the previous frame.

Transmit Flow Control

In full-duplex mode, when the Transmit Flow Control Enable bit ([TFCE](#)) is set to 1, the MAC generates pause frames and transmits them as necessary, with the calculated CRC appended.

Pause frame generation can be initiated in two ways. When the application sets the Flow Control Busy bit ([FCBBA](#)) to 1, or when the RX FIFO is full, a pause frame is transmitted.

- If the application has requested flow control by setting the [FCBBA](#) bit to 1, the MAC will generate and transmit a single pause frame. The value of the pause time in the generated frame is the pause time value programmed in the [PAUSE_TIME](#) field. If the application wants to extend or end the pause prior to the time specified in the previously transmitted pause frame, it must request another pause frame transmission after programming the pause time field [PAUSE_TIME](#) with an appropriate value.
- If the application has requested flow control when the RX FIFO is full, the MAC will generate and transmit a pause frame. The value of the pause time in the generated frame is the pause time value programmed in the [PAUSE_TIME](#) field. If the RX FIFO remains full at a configurable number of slot times ([PLT](#)) before the pause time runs out, a second pause frame will be transmitted. The process will be repeated as long as the RX FIFO remains full. If the FIFO is no longer full prior to the sampling time, the MAC will send a pause frame with zero pause time to indicate to the remote end that the RX buffer is ready to receive new data frames.

Retransmission During Collision

In half-duplex mode, a collision may occur on the MAC line interface when frames are transmitted to the MAC. The MAC may indicate a retry attempt by giving the status even before the EOF is transferred. Then the MAC will enable the retransmission by popping out the frame again from the FIFO. After more than 96 bytes are popped towards the MAC core, the FIFO controller frees the space in the FIFO, and makes it available to the DMA to push in more data. This means that the retransmission is not possible after this threshold is crossed or when the MAC core indicates a late-collision event.

The MAC transmitter may abort the transmission of a frame because of collision, TX FIFO underflow, loss of carrier, jabber timeout, no carrier, excessive deferral, or late collision. When frame transmission is aborted because of collision, the MAC requests retransmission of the frame.

Transmit Status Word

At the end of the Ethernet frame transfer, the MAC outputs the transmit status to the application. The detailed description of the Transmit Status is the same as for [TDES0](#).

52.4.1.2 Reception

A receive operation is initiated when the MAC detects an SFD on the RMII or MII. The MAC strips the preamble and SFD before proceeding to process the frame. The header fields are checked for filtering and the FCS (Frame Check Sequence) field used to verify the CRC for the frame. The received frame is stored in a shallow buffer until the address filtering is performed. The frame is dropped in the MAC if it fails the address filtering.

The frames received by the MAC are pushed into the RX FIFO. Once the RX FIFO crosses the configured Receive threshold ([RX_THRESH_CTRL](#)), its status is indicated to the DMA, so that the DMA can initiate a pre-configured burst transfers towards the AHB interface.

In the default cut-through mode, when the FIFO receives 64 bytes or a full packet of data, it pops out the data and indicates the availability to the DMA. Once the DMA initiates the transfer to the AHB interface, the data transmission continues from the FIFO until a complete packet has been transferred. Upon the completion of the EOF frame transfer, the status word will be popped out and transmitted to the DMA controller.

Receive Protocol

The receive module strips the preamble and SFD of the received frame. Once the SFD is detected, the MAC begins sending the Ethernet frame data to the RX FIFO, starting from the first byte following the SFD (destination address). If the IEEE 1588 timestamp feature is enabled, the snapshot of system time will be captured whenever an SFD of any frame is detected on MII, and sent to the application unless the MAC filters out and drops the frame.

If the Length/Type field of the receiving frame is less than 0x600 and if the MAC is programmed for the automatic CRC/paddingstripping option, the MAC sends the frame data up to the count specified in the Length/Type field to the RX FIFO, and then starts dropping bytes (including the FCS field). If the Length/Type field is greater than or equal to 0x600, the MAC will send all received Ethernet frame data to the RX FIFO, regardless of the value on the programmed automatic CRC stripping option. By default, the MAC is programmed for the watchdog timer to be enabled, that is frames above 2048 bytes (including DA, SA, LT, data, padding, and FCS) are cut off. This feature can be disabled by programming the Watchdog Disable bit [EMACWATCHDOG](#). However, even if the watchdog timer is disabled, frames longer than 16 KB will still be cut off and the watchdog timeout status will be indicated.

Receive Frame Controller

If the [RECEIVE_ALL](#) bit in the MAC Frame Filter Register is reset, the MAC performs frame filtering based on the destination and source addresses. The application still needs to perform another level of filtering if it decides not to receive any bad frames like runt, CRC error frames, etc. On detecting a filter failure, the frame is dropped and not transmitted to the application. When the filter parameters change dynamically, if a DA and SA filter failure occurs, the reset of the frame is dropped and the Receive Status word (with zero frame length, CRC Error, and Runt Error bits set) is updated immediately indicating the filter failure.

Receive Flow Control

The MAC detects the receiving pause frame and pauses the frame transmission for the delay specified within the received pause frame (in full-duplex mode only). The Pause Frame Detection Function can be enabled or disabled with the [RFCE](#) bit. Once the receive flow control is enabled, the MAC starts monitoring the received frame destination address for any match with the multicast address of the control frame (0x0180 C200 0001).

If a match is detected (i.e., the destination address of the received frame matches the destination address of the reserved control frame), the MAC decides whether to transmit the received control frame to the application, according to the [PCF](#) field.

The MAC also decodes the type, the opcode, and the pause timer field of the receiving control Frame. If the byte count of the status indicates 64 bytes, and if there is no CRC error, the MAC transmitter will pause the transmission of any data frame for the duration of the decoded pause time value multiplied by the slot time (64 byte times for 10/100 Mbit/s mode). Meanwhile, if another pause frame is detected with a zero pause time, the MAC will reset the pause time and give another pause request.

If the received control frame matches neither the type field (0x8808), the opcode (0x00001), nor the byte length (64 bytes), or if there is a CRC error, the MAC will not generate a pause.

In the case of a pause frame with a multicast destination address, the MAC filters the frame according to the address match.

For a pause frame with a unicast destination address, the filtering depends on whether the DA matched the contents of the MAC Address Register 0 and the [UPFD](#) (detecting a pause frame even with a unicast destination address) is set. The [PCF](#) register controls the filtering for control frames in addition to the address filter.

Receive Operation Multiframe Handling

Since the status is available immediately following the data, the MAC is capable of storing any number of frames into the FIFO, as long as it is not full.

Error Handling

If the RX FIFO is full before it receives the EOF data from the MAC, an overflow will be declared and the whole frame will be dropped. The status bit [RDESO](#)[11] will reflect the fact that this frame is a partial frame due to overflow. The RX FIFO can perform the filtering of error frames and runt frames, if this function is enabled via [FLUSH_TX_FIFO](#) and [FWD_UNDER_GF](#).

In cut-through mode, if a frame's status and length are available when reading an SOF from the RX FIFO, the whole error frame can be dropped. The DMA can flush the error frame being read from the FIFO by clearing [FWD_ERR_FRAME](#) to 0. The DMA then stops transferring data to the application, internally reads out the rest of the frame, and drops it. The MTL will then start the transfer of the next frame, if it is available. If FIFO is available, the transmission of the next frame will be initiated.

Receive Status Word

At the end of the Ethernet frame transfer, the MAC outputs the receive status to the application via [EMAC_DMA](#). The detailed description of the receive status is the same for bit[31:0] in [RDESO](#).

52.4.2 EMAC_MTL (MAC Transaction Layer)

The MAC Transaction Layer provides FIFO memory to buffer and regulate the frames between the application system memory and the MAC. It also enables the data to be transmitted between the application clock domain and the MAC clock domains. The MTL layer has two data paths, namely the Transmit path and the Receive path. The data path for both directions is 32-bit wide and operates with a simple FIFO protocol.

52.4.3 EMAC_DMA

EMAC has its own DMA module, with independent Transmit and Receive engines and a CSR (Control and Status Registers) space. The Transmit engine transfers data from the system memory to the device port (MTL), while the Receive engine transmits data from the device port to the system memory. The controller uses descriptors to efficiently move data from source to destination with minimal Host CPU intervention. The DMA is designed for packet-oriented data transfer such as frames in Ethernet. The controller can be programmed to interrupt the Host CPU for normal situations, such as the completion of frame transmission or reception, or error conditions.

52.4.3.1 Transmit Descriptors

Figure 52.4-1 shows the structure of the transmit descriptor linked list. Table 52.4-1 through Table 52.4-4 show the descriptions of each transmit descriptor.

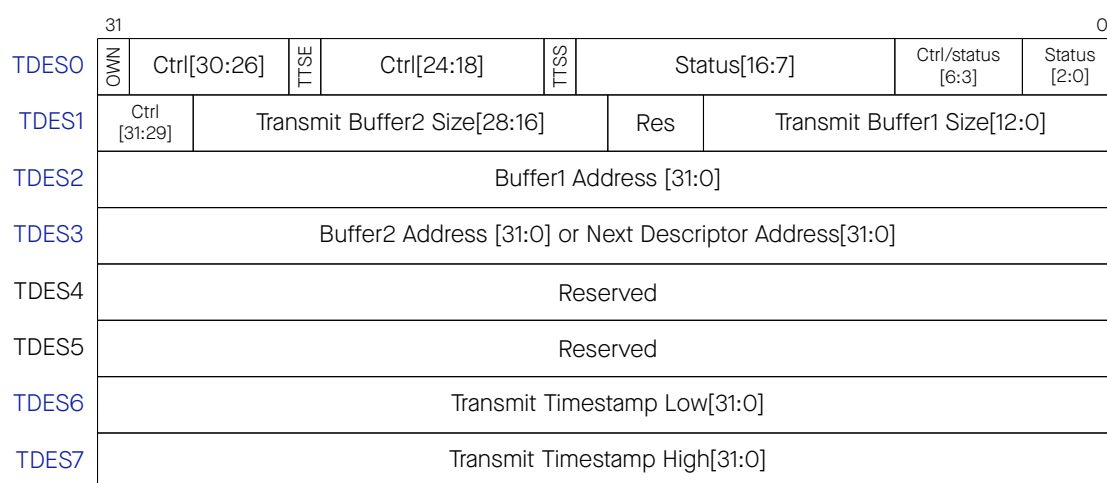


Figure 52.4-1. Transmit Descriptor Linked List

Table 52.4-1. Transmit Descriptor 0 (TDES0)

Bits	Name	Description
[31]	OWN: Own Bit	When set, this bit indicates that the descriptor is owned by the DMA. When reset, it indicates that the descriptor is owned by the Host. The DMA clears this bit, either when it completes the frame transmission or when the buffers allocated to the descriptor are empty. The ownership bit of the First Descriptor of the frame should be set after all subsequent descriptors belonging to the same frame have been set. This avoids a possible race condition between fetching a descriptor and the driver setting an ownership bit.
[30]	IC: Interrupt on Completion	When set, this bit sets the Transmit Interrupt TRANS_INT after the present frame has been transmitted. This bit is valid only when the last segment bit TDES0[29] is set.
[29]	LS: Last Segment	When set, this bit indicates that the buffer contains the last segment of the frame, and the TBS1 or TBS2 field in TDES1 should have a non-zero value.

Bits	Name	Description
[28]	FS: First Segment	When set, this bit indicates that the buffer contains the first segment of a frame.
[27]	DC: Disable CRC	When this bit is set, the MAC does not append a cyclic redundancy check (CRC) to the end of the transmitted frame. This is valid only when the first segment (TDES0[28]) is set.
[26]	DP: Disable Padding	When this bit is set, the MAC does not automatically add padding to a frame shorter than 64 bytes. When this bit is reset, the DMA automatically adds padding and CRC to a frame shorter than 64 bytes, and the CRC field is added despite the state of the DC (TDES0[27]) bit. This is valid only when the first segment (TDES0[28]) is set.
[25]	TTSE: Transmit Time Stamp Enable	When this bit is set, the MAC enables IEEE1588 hardware timestamping for the transmit frame.
[24]	CRCR: CRC Replacement Control	When set, the MAC replaces the last four bytes of the transmitted packet with recalculated CRC bytes. The Host should ensure that the CRC bytes are present in the frame being transmitted from the Transmit Buffer. This bit is valid when the First Segment control bit (TDES0[28]) is set. In addition, CRC replacement is done only when Bit TDES0[27] is set to 1.
[23:22]	CIC: Checksum Insertion Control	These bits control the checksum calculation and insertion. Bit encodings are shown below: • 00: Checksum insertion is disabled. • 01: Only IP header checksum calculation and insertion are enabled. • 10: IP header checksum and payload checksum calculation and insertion are enabled, but pseudo-header checksum is not calculated in hardware. • 11: IP Header checksum and payload checksum calculation and insertion are enabled, and pseudo-header checksum is calculated in hardware. This field is valid when the First Segment control bit (TDES0[28]) is set.
[21]	TER: Transmit End of Ring	When set, this bit indicates that the descriptor list reached its final descriptor. The DMA returns to the base address of the list, creating a Descriptor Ring.
[20]	TCH: Second Address Chained	When set, this bit indicates that the second address in the descriptor is the Next Descriptor address, rather than the second buffer address. When TDES0[20] is set, TBS2 (TDES1[28:16]) is a “don’t care” value. TDES0[21] takes precedence over TDES0[20]. This bit must be set to 1.

Bits	Name	Description
[19:18]	VLIC: VLAN Insertion Control	When set, these bits request the MAC to perform VLAN tagging or untagging before transmitting the frames. If the frame is modified for VLAN tags, the MAC automatically recalculates and replaces the CRC bytes. The following list describes the values of these bits: • 00: Do not add a VLAN tag. • 01: Remove the VLAN tag from the frames before transmission. This option should be used only with the VLAN frames. • 10: Insert a VLAN tag with the tag value programmed in VLAN Tag Inclusion or Replacement Register. • 11: Replace the VLAN tag in frames with the Tag value programmed in VLAN Tag Inclusion or Replacement Register. This option should be used only with the VLAN frames.
[17]	TTSS: Transmit Timestamp Status	This bit is used as a status bit to indicate that a timestamp was captured for the described transmit frame. When this bit is set, TDES6 and TDES7 have a timestamp value captured for the transmit frame. This bit is valid only when the descriptor's LS (TDES0[29]) is set.
[16]	IHE: IP Header Error	When set, this bit indicates that the MAC transmitter detected an error in the IP datagram header. The transmitter checks the header length in the IPv4 packet against the number of header bytes received from the application, and indicates an error status if there is a mismatch. For IPv6 frames, a header error is reported if the main header length is not 40 bytes. Furthermore, the Ethernet Length/Type field value for an IPv4 or IPv6 frame must match the IP header version received with the packet. For IPv4 frames, an error status is also indicated if the Header Length field has a value less than 0x5.
[15]	ES: Error Summary	Indicates the logical OR of the following bits: • TDES0[16]: IP Header Error • TDES0[14]: Jabber Timeout • TDES0[13]: Frame Flush • TDES0[12]: IP Payload Error • TDES0[11]: Loss of Carrier • TDES0[10]: No Carrier • TDES0[9]: Late Collision • TDES0[8]: Excessive Collision • TDES0[2]: Excessive Deferral • TDES0[1]: Underflow Error
[14]	JT: Jabber Timeout	When set, this bit indicates the MAC transmitter has experienced a jabber timeout. This bit is only set when the EMACJABBER bit (disable jabber) of EMACCONFIG_REG is not set.
[13]	FF: Frame Flushed	When set, this bit indicates that the DMA or MTL flushed the frame because of a software Flush command given by the CPU.

Bits	Name	Description
[12]	IPE: IP Payload Error	When set, this bit indicates that the MAC transmitter detected an error in the TCP, UDP, or ICMP IP datagram payload. The transmitter checks the payload length received in the IPv4 or IPv6 header against the actual number of TCP, UDP, or ICMP packet bytes received from the application, and issues an error status in case of a mismatch.
[11]	LOC: Loss of Carrier	When set, this bit indicates that a Loss of Carrier occurred during frame transmission (that is, the MII_CRS signal was inactive for one or more transmit clock periods during frame transmission). This is valid only for the frames transmitted without collision and when the MAC operates in the half-duplex mode.
[10]	NC: No Carrier	When set, this bit indicates that the carrier sense signal from the PHY was not asserted during transmission.
[9]	LC: Late Collision	When set, this bit indicates that frame transmission was aborted because of a collision occurring after the collision window (in MII mode, 64 byte times including Preamble, and 512 byte times including Preamble and Carrier Extension). This bit is not valid if the Underflow Error bit is set.
[8]	EC: Excessive Collision	When set, this bit indicates that the transmission was aborted after 16 successive collisions while attempting to transmit the current frame. If the EMACRETRY bit of EMACCONFIG_REG is set, this bit is set after the first collision, and the transmission of the frame is aborted.
[7]	VF: VLAN Frame	When set, this bit indicates that the transmitted frame was a VLAN-type frame.
[6:3]	Ctrl/status	These status bits indicate the number of collisions occurring before the frame was transmitted. This count is not valid when the Excessive Collisions bit (TDES0[8]) is set. The CPU updates this status field only in the half-duplex mode.
[2]	ED: Excessive Deferral	When set, this bit indicates that the transmission has ended because of excessive deferral of over 24,288 bit times (if Jumbo Frame is enabled) if the EMACDEFERRAL bit (deferral check) of EMACCONFIG_REG is set high.
[1]	UF: Underflow Error	When set, this bit indicates that the MAC aborted the frame because the data arrived late from the Host memory. Underflow Error indicates that the DMA encountered an empty transmit buffer while transmitting the frame. The transmission process enters the Suspended state and sets both Transmit Underflow Register (Status Register TRANS_UNDFLOW) and Transmit Interrupt Register (Status Register TRANS_INT).
[0]	DB: Deferred Bit	When set, this bit indicates that the MAC defers before transmission because of the presence of a carrier. This bit is valid only in the half-duplex mode.

Table 52.4-2. Transmit Descriptor 1 (TDES1)

Bits	Name	Description
[31:29]	SAIC: SA Insertion Control	These bits request the MAC to add or replace the Source Address field in the Ethernet frame with the value given in EMACADDROHIGH_REG, EMACADDROLOW_REG, EMACADDRHIGH_REG, and EMACADDRLOW_REG. If the Source Address field is modified in a frame, the MAC automatically recalculates and replaces the CRC bytes. The Bit[31] specifies the MAC Address Register value (1 or 0) that is used for Source Address insertion or replacement. The following list describes the values of Bits[30:29]: • 00: Do not include the source address. • 01: Include or insert the source address. For reliable transmission, the application must provide frames without source addresses. • 10: Replace the source address. For reliable transmission, the application must provide frames with source addresses. • 11: Reserved. These bits are valid when the First Segment control bit (TDES0[28]) is set.
[28:16]	TBS2: Transmit Buffer2 Size	These bits specify the transmit Buffer 2 size in bytes. This field is not valid if TDES0[20] is set.
[15:13]	Reserved	Reserved
[12:0]	TBS1: Transmit Buffer 1 Size	These bits indicate the data buffer byte size in bytes. If this field is 0, the DMA ignores this buffer and uses Buffer 2 or the next descriptor, depending on the value of TCH (TDES0[20]).

Table 52.4-3. Transmit Descriptor 2 (TDES2)

Bits	Name	Description
[31:0]	Buffer 1 Address Pointer	These bits indicate the physical address of Buffer 1.

Table 52.4-4. Transmit Descriptor 3 (TDES3)

Bits	Name	Description
[31:0]	Buffer2 Address Pointer (Next Descriptor Address)	These bits specify the physical address of Buffer 2 when a descriptor ring structure is used. If the Second Address Chained (TDES0[20]) bit is set, this address specifies the physical memory where the Next Descriptor is present.

Table 52.4-5. Transmit Descriptor 6 (TDES6)

Bits	Name	Description
[31:0]	TTSL: Transmit Frame Time Stamp Low	This field is updated by DMA with the least significant 32 bits of the timestamp captured for the corresponding transmit frame. This field has the timestamp only if the Last Segment (TDES0[29]) bit in the descriptor is set, and the Timestamp Status (TDES0[17]) bit is set.

Table 52.4-6. Transmit Descriptor 7 (TDES7)

Bits	Name	Description
[31:0]	TTSH: Transmit Frame Time Stamp High	This field is updated by DMA with the most significant 32 bits of the timestamp captured for the corresponding receive frame. This field has the timestamp only if the Last Segment (TDES0[29]) bit in the descriptor is set, and the Timestamp Status (TDES0[17]) bit is set.

52.4.3.2 Receive Descriptors

Figure 52.4-2 shows the structure of the receive descriptor linked list. Table 52.4-7 through Table 52.4-11 show the descriptions of each receive descriptor.

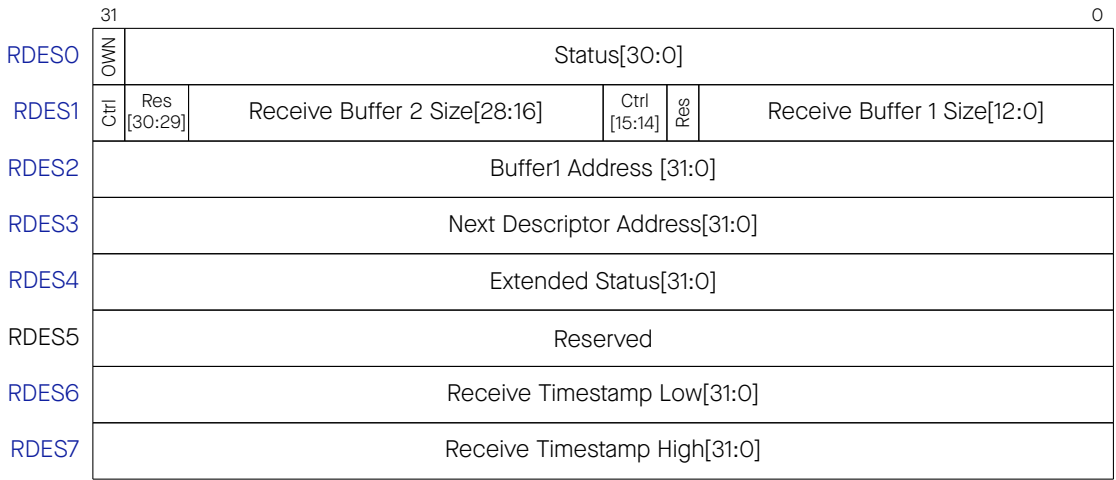


Figure 52.4-2. Receive Descriptor Linked List

Table 52.4-7. Receive Descriptor 0 (RDES0)

Bits	Name	Description
[31]	OWN: Own Bit	When set, this bit indicates that the descriptor is owned by the DMA. When reset, it indicates that the descriptor is owned by the Host. The DMA clears this bit either when it completes the frame reception or when the buffers that are associated with this descriptor are full.

Bits	Name	Description
[30]	AFM: Destination Address Filter Fail	When set, this bit indicates a frame that failed in the DA Filter in the MAC.
[29:16]	FL: Frame Length	These bits indicate the byte length of the received frame that was transmitted to Host memory. This field is valid when Last Descriptor (RDESO[8]) is set and either the Descriptor Error (RDESO[14]) or Overflow Error bit RDESO[11] are reset. The frame length also includes the two bytes appended to the Ethernet frame when IP checksum calculation (Type 1) is enabled and the received frame is not a MAC control frame.
[15]	ES: Error Summary	Indicates the logical OR of the following bits: • RDESO[1]: CRC Error • RDESO[3]: Receive Error • RDESO[4]: Watchdog Timeout • RDESO[6]: Late Collision • RDESO[7]: Giant Frame • RDES4[4:3]: IP Header or Payload Error • RDESO[11]: Overflow Error • RDESO[14]: Descriptor Error This field is valid only when RDESO[8] is set.
[14]	DE: Descriptor Error	When set, this bit indicates a frame truncation caused by a frame that does not fit within the current descriptor buffers, and that the DMA does not own the Next Descriptor. The frame is truncated. This field is valid only when This field is valid only when RDESO[8] is set.
[13]	SAF: Source Address Filter Fail	When set, this bit indicates that the SA field of frame failed the SA Filter in the MAC.
[12]	LE: Length Error	When set, this bit indicates that the actual length of the frame received and that the Length/Type field does not match. This bit is valid only when the Frame Type (RDESO[5]) bit is reset.
[11]	OE: Overflow Error	When set, this bit indicates that the received frame was damaged because of buffer overflow in MTL.
[10]	VLAN: VLAN Tag	When set, this bit indicates that the frame pointed to by this descriptor is a VLAN frame tagged by the MAC. The VLAN tagging depends on checking the VLAN fields of the received frame based on the VLAN Tag Register) configuration.
[9]	FS: First Descriptor	When set, this bit indicates that this descriptor contains the first buffer of the frame. If the size of the first buffer is 0, the second buffer contains the beginning of the frame. If the size of the second buffer is also 0, the next Descriptor contains the beginning of the frame.
[8]	LS: Last Descriptor	When set, this bit indicates that the buffers pointed to by this descriptor are the last buffers of the frame.

Bits	Name	Description
[7]	IP Checksum Error (Type1), or Giant Frame	When the Advanced Timestamp feature is present, and when this bit set, it indicates that a snapshot of the Timestamp is written in descriptor word 6 (RDES6) and 7 (RDES7). This is valid only when the Last Descriptor bit (RDES0[8]) is set. When IP Checksum Engine (Type 1) is selected, this bit, if set, indicates one of the following: • The 16-bit IPv4 header checksum calculated by the core did not match the received checksum bytes. • The header checksum checking is bypassed for non-IPv4 frames. Otherwise, this bit, when set, indicates the Giant Frame Status. When normal frame processing is enabled, Giant frames are larger than 1,518 bytes (or 1,522 bytes for VLAN or 2,000 bytes when ASS2KP is set). When Jumbo Frame processing is enabled, Giant frames are larger than 9,018 bytes (or 9,022 bytes for VLAN).
[6]	LC: Late Collision	When set, this bit indicates that a late collision has occurred while receiving the frame in the half-duplex mode.
[5]	FT: Frame Type	When set, this bit indicates that the Receive Frame is an Ethernet-type frame (the LT field is greater than, or equal to, 1,536 bytes). When reset, it indicates that the received frame is an IEEE 802.3 frame. This bit is not valid for Runt frames which are less than 14 bytes.
[4]	RWT: Receive Watchdog Timeout	When set, this bit indicates that the Receive Watchdog Timer has expired while receiving the current frame and the current frame is truncated after the Watchdog Timeout.
[3]	RE: Receive Error	When set, this bit indicates that the MII_RXER signal is asserted while MII_RXDV is asserted during frame reception.
[2]	DE: Dribble Bit Error	When set, this bit indicates that the received frame has a non-integer multiple of bytes (odd nibbles). This bit is valid only in the MII Mode.
[1]	CE: CRC Error	When set, this bit indicates that a Cyclic Redundancy Check (CRC) Error occurred on the received frame. This field is valid only when the Last Descriptor (RDES0[8]) is set.

Bits	Name	Description
[0]	Extended Status Available/ RX MAC Address	When the IP Checksum Offload (Type 2) is present, this bit, when set, indicates that the extended status is available in descriptor word 4 (RDES4). This is valid only when the Last Descriptor bit (RDES0[8]) is set. This bit is invalid when Bit 30 is set. When IP Checksum Offload (Type 2) is present, this bit is set even when the IP Checksum Offload engine bypasses the processing of the received frame. The bypassing may be because of a non-IP frame or an IP frame with a non-TCP/UDP/ICMP payload. When the IPC Full Offload is not selected, this bit indicates an RX MAC Address status. When set, this bit indicates that the RX MAC Address registers value (1 to 15) matched the frame's DA field. When reset, this bit indicates that the RX MAC Address Register 0 value matched the DA field.

Table 52.4-8. Receive Descriptor 1 (RDES1)

Bits	Name	Description
[31]	Ctrl	When set, this bit prevents setting the Status Register's RI bit (RECV_INT) for the received frame that ends in the buffer indicated by this descriptor. This, in turn, disables the assertion of the interrupt to Host because of the RI for that frame.
[30:29]	Reserved	Reserved.
[28:16]	RBS2: Receive Buffer 2 Size	These bits indicate the size of Buffer 2 in bytes. The buffer size must be a multiple of 4, even if the value of RDES3 (Buffer 2 address pointer) is not aligned to bus width. In the case where the buffer size is not a multiple of 4, the resulting behavior is undefined. This field is not valid if RDES1[14] is set.
[15]	RER: Receive End of Ring	When set, this bit indicates that the descriptor list reached its final descriptor. The DMA returns to the base address of the list, creating a Descriptor Ring.
[14]	RCH: Second Address Chained	When set, this bit indicates that the second address in the descriptor is the Next Descriptor address rather than the second buffer address. When this bit is set, RBS2 (RDES1[28:16]) is a "don't care" value. RDES1[15] takes precedence over RDES1[14] .
[13]	Reserved	Reserved.
[12:0]	RBS1: Receive Buffer 1 Size	These bits indicate the size of Buffer 1 in bytes. The buffer size must be a multiple of 4, even if the value of RDES2 (buffer1 address pointer) is not aligned to bus width. When the buffer size is not a multiple of 4, the resulting behavior is undefined. If this field is 0, the DMA ignores this buffer and uses Buffer 2 or the next descriptor depending on the value of RCH (Bit[14]).

Table 52.4-9. Receive Descriptor 2 (RDES2)

Bits	Name	Description
[31:0]	Buffer 1 Address Pointer	These bits indicate the physical address of Buffer 1.

Table 52.4-10. Receive Descriptor 3 (RDES3)

Bits	Name	Description
[31:0]	Buffer 2 Address Pointer (Next Descriptor Address)	These bits indicate the physical address of Buffer 2 when descriptor chaining is used. If the Second Address Chained (RDES1[14]) bit is set, then this address contains the pointer to the physical memory where the Next Descriptor is present.

Table 52.4-11. Receive Descriptor 4 (RDES4)

Bits	Name	Description
[31:28]	Reserved	Reserved.
[27:26]	Reserved	Reserved.
[25]	Reserved	Reserved.
[24]	Reserved	Reserved.
[23:21]	Reserved	Reserved.
[20:18]	Reserved	Reserved.
[17]	Reserved	Reserved.
[16]	Reserved	Reserved.
[15]	Reserved	Reserved.
[14]	Timestamp Dropped	When set, this bit indicates that the timestamp was captured for the receive frame but got dropped in the MTL RX FIFO because of overflow.
[13]	PTP Version	When set, this bit indicates that the received PTP message is having the IEEE 1588 version 2 format. When reset, it has the version 1 format.
[12]	PTP Frame Type	When set, this bit indicates that the PTP message over Ethernet is sent directly over Ethernet. When this bit is not set and the MT (message type) is not 0, it indicates that the PTP message over UDP-IPv4 or UDP-IPv6 is received.

Bits	Name	Description
[11:8]	Message Type	These bits are encoded to give the type of the message received. • 0000: Reserved • 0001: SYNC (all clock types) • 0010: Follow_Up (all clock types) • 0011: Delay_Req (all clock types) • 0100: Delay_Resp (all clock types) • 0101: Pdelay_Req (in peer-to-peer transparent clock) • 0110: Pdelay_Resp (in peer-to-peer transparent clock) • 0111: Pdelay_Resp_Follow_Up (in peer-to-peer transparent clock) • 1000: Announce • 1001: Management • 1010: Signaling • 1011-1110: Reserved • 1111: PTP packet with Reserved message type
[7]	IPv6 Packet Received	When set, this bit indicates that the received packet is an IPv6 packet. This bit is updated only when Bit[10] (IPC) of MAC Configuration Register (EMACRXIPCOFFLOAD) is set.
[6]	IPv4 Packet Received	When set, this bit indicates that the received packet is an IPv4 packet. This bit is updated only when Bit[10] (IPC) of MAC Configuration Register (EMACRXIPCOFFLOAD) is set.
[5]	IP Checksum Bypassed	When set, this bit indicates that the checksum offload engine is bypassed.
[4]	IP Payload Error	When set, this bit indicates that the 16-bit IP payload checksum (that is, the TCP, UDP, or ICMP checksum) that the MAC core calculated does not match the corresponding checksum field in the received segment. It is also set when the TCP, UDP, or ICMP segment length does not match the payload length value in the IP Header field. This bit is valid when either Bit[7] or Bit[6] is set.
[3]	IP Header Error	When set, this bit indicates that either the 16-bit IPv4 header checksum calculated by the MAC core does not match the received checksum bytes, or the IP datagram version is not consistent with the Ethernet Type value. This bit is valid when either Bit[7] or Bit[6] is set.

Bits	Name	Description
[2:0]	IP Payload Type	These bits indicate the type of payload encapsulated in the IP datagram processed by the Receive Checksum Offload Engine (COE). The COE also sets these bits to 00 if it does not process the IP datagram's payload due to an IP header error or fragmented IP. • 000: Unknown or did not process IP payload • 001: UDP • 010: TCP • 011: ICMP • 1xx: Reserved This bit is valid when either Bit[7] or Bit[6] is set.

Table 52.4-12. Receive Descriptor 6 (RDES6)

Bits	Name	Description
[31:0]	RTSH: Receive Frame Time Stamp Low	This field is updated by DMA with the least significant 32 bits of the timestamp captured for the corresponding receive frame. This field is updated by DMA only for the last descriptor of the receive frame which is indicated by the Last Descriptor status bit (RDES0[8]).

Table 52.4-13. Receive Descriptor 7 (RDES7)

Bits	Name	Description
[31:0]	RTSH: Receive Frame Timestamp High	This field is updated by DMA with the most significant 32 bits of the timestamp captured for the corresponding receive frame. This field is updated by DMA only for the last descriptor of the receive frame which is indicated by the Last Descriptor status bit (RDES0[8]).

52.4.4 PHY Interface

The MAC and the external PHY communicate through two interfaces:

- Media Independent Interface (MII)
- Reduced Media Independent Interface (RMII)

52.4.4.1 Media Independent Interface: MII

The Media Independent Interface (MII) defines the interconnection between MAC sublayers and PHYs at 10 Mbit/s and 100 Mbit/s.

MII Signals to PHY

The MII interface signals are shown in Figure [52.4-3](#).

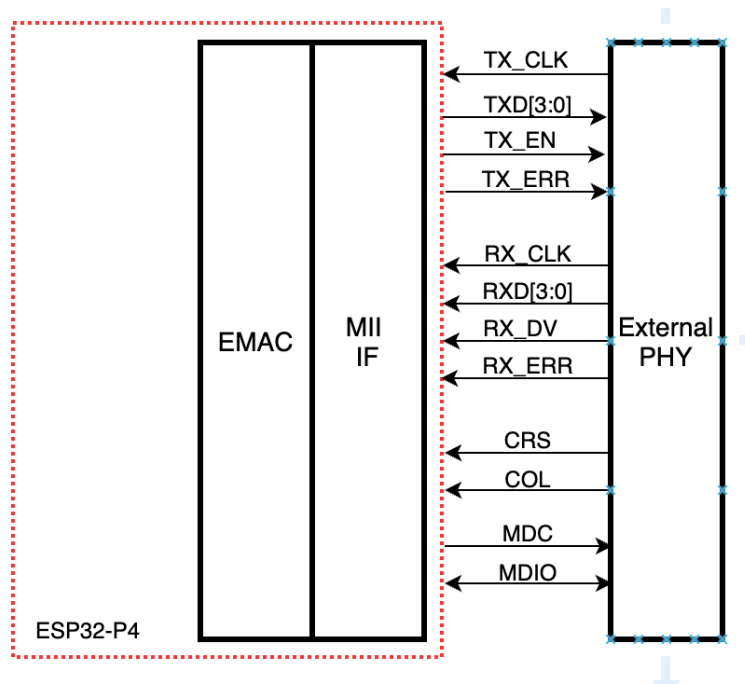


Figure 52.4-3. MII Interface

The MII interface signals are described below:

- **MII_TX_CLK**: TX clock signal. This signal provides the reference timing for TX data transmission. It has two frequencies: 2.5 MHz at 10 Mbit/s, and 25 MHz at 100 Mbit/s.
- **MII_TXD[3:0]**: A bundle of four transmit data signals driven by the MAC sublayer, and qualified (valid data) on the assertion of the MII_TX_EN signal. MII_TXD[0] is the lowest significant bit and MII_TXD[3] is the highest significant bit. When the MII_TX_EN signal is deasserted, the transmit data have no effect on the PHY.
- **MII_TX_EN**: Transmission enable signal. This signal indicates that the MAC is presenting nibbles (4 bits) on the MII for transmission. It must be asserted synchronously (MII_TX_CLK) with the first nibble of the preamble and must remain asserted while all nibbles to be transmitted are presented to the MII.
- **MII_TX_ERR**: Transmit error signal. The signal must be asserted for one or more clock periods (MII_TX_CLK) to indicate to the PHY layer that an error was detected somewhere in the frame.
- **MII_RX_CLK**: RX clock signal. This signal provides the reference timing for RX data transmission. It has two frequencies: 2.5 MHz at 10 Mbit/s, and 25 MHz at 100 Mbit/s.
- **MII_RXD[3:0]**: A bundle of four receive data signals driven by the MAC sublayer, and qualified (valid data) on the assertion of the MII_RX_DV signal. MII_RXD[0] is the lowest significant bit and MII_RXD[3] is the highest significant bit. When MII_RX_DV is deasserted and MII_RX_ERR is asserted, a specific MII_RXD[3:0] value is used to transfer specific information from the PHY.
- **MII_RX_DV**: Receive data valid signal. This signal indicates that the PHY is presenting recovered and decoded nibbles on the MII for reception. It must be asserted synchronously (MII_RX_CLK) with the first recovered nibble of the frame and must remain asserted through the final recovered nibble. It must be deasserted prior to the first clock cycle that follows the final nibble. In order to receive the frame correctly, the MII_RX_DV signal must encompass the frame, starting no later than the SFD field.

- **MII_CRS:** Carrier sense signal. When the transmit or receive medium is non-idle, the signal is asserted by the PHY. When the transmit or receive medium is idle, the signal is deasserted by the PHY. The PHY must ensure that the MII_CRS signal remains asserted throughout the duration of a collision condition. This signal does not need to be synchronized with the TX and RX clocks. In full duplex mode the state of this signal is don't care for the MAC sublayer.
- **MII_COL:** Collision detection signal. This signal must be asserted by the PHY upon detection of a collision on the medium and must remain asserted while the collision condition persists. This signal is not required to transition synchronously with respect to the TX and RX clocks. In full duplex mode, the state of this signal is don't care for the MAC sublayer.
- **MII_RX_ERR:** Receive error signal. The signal must be asserted for one or more clock periods (MII_RX_CLK) to indicate to the MAC sublayer that an error was detected somewhere in the frame.
- **MDIO and MDC:** Management data input/output and management data clock signal. The two signals constitute a serial bus defined for the Ethernet family of IEEE 802.3 standards, used to transfer control and data information to the PHY, see section [Station Management Agent \(SMA\) Interface](#).

MII Clock

In MII mode, there are clock signals in two directions between MII and the PHY, namely `clk_tx` and `clk_rx`. `clk_tx` is used to synchronize the TX data, and `clk_rx` is used to synchronize the RX data. Both clocks are provided by the PHY.

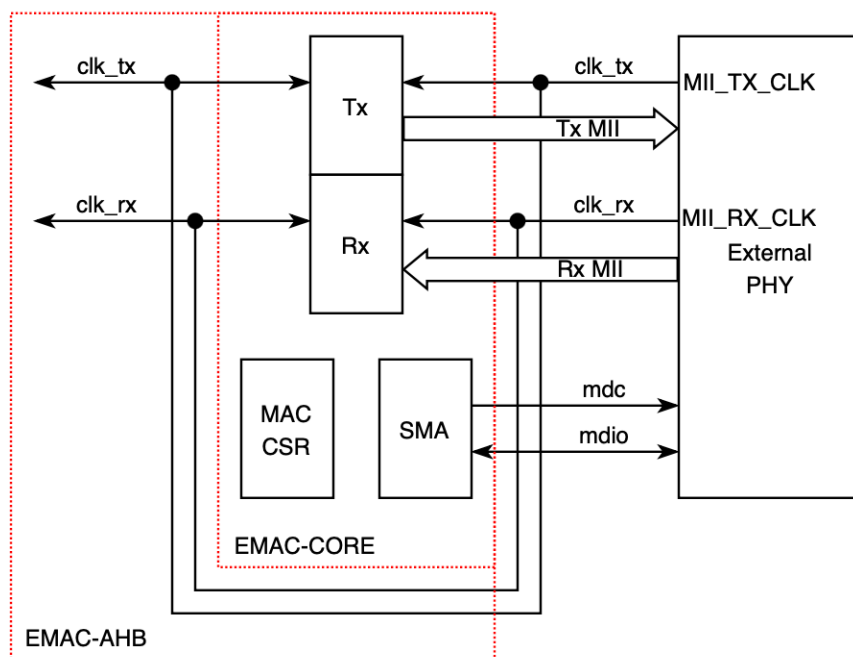


Figure 52.4-4. MII Clock

52.4.4.2 Reduced Media Independent Interface: RMII

RMII interface signals are shown in Figure [52.4-5](#).

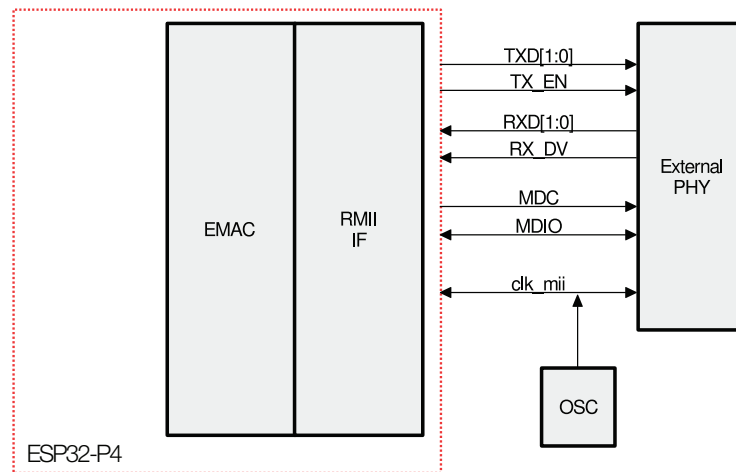


Figure 52.4-5. RMII Interface

Note:

Figure 52.4-3 and Figure 52.4-5 uniformly use the signal name RX_DV to represent the receive-valid signal. In the MII interface, this corresponds to RX_DV, while in the RMII interface it corresponds to CRS_DV. The actual signal semantics depend on the selected Ethernet interface mode.

RMII Signals

The Reduced Media Independent Interface (RMII) specification reduces the number of pins between the microcontroller's Ethernet MAC and the external PHY at 10 Mbit/s or 100 Mbit/s. According to the IEEE 802.3u standard, an MII contains 16 pins for data and control. The RMII specification reduces the pin count to 7.

RMII has the following features:

- Supports an operating rate of 10 Mbit/s or 100 Mbit/s
- The clock reference must be 50 MHz.
- The same clock reference must be sourced externally both to the MAC and the external Ethernet PHY.
- Provides independent 2-bit wide TX and RX data paths.

RMII Clock

The 50 MHz RMII reference clock can be:

- Generated internally by ESP32-P4 and looped back via GPIOs. For GPIO configurations, please refer to Chapter 9 *GPIO Matrix and IO MUX*.
- Provided by the external crystal and obtained at GPIOs.

The RMII clock is shown in Figure 52.4-6.

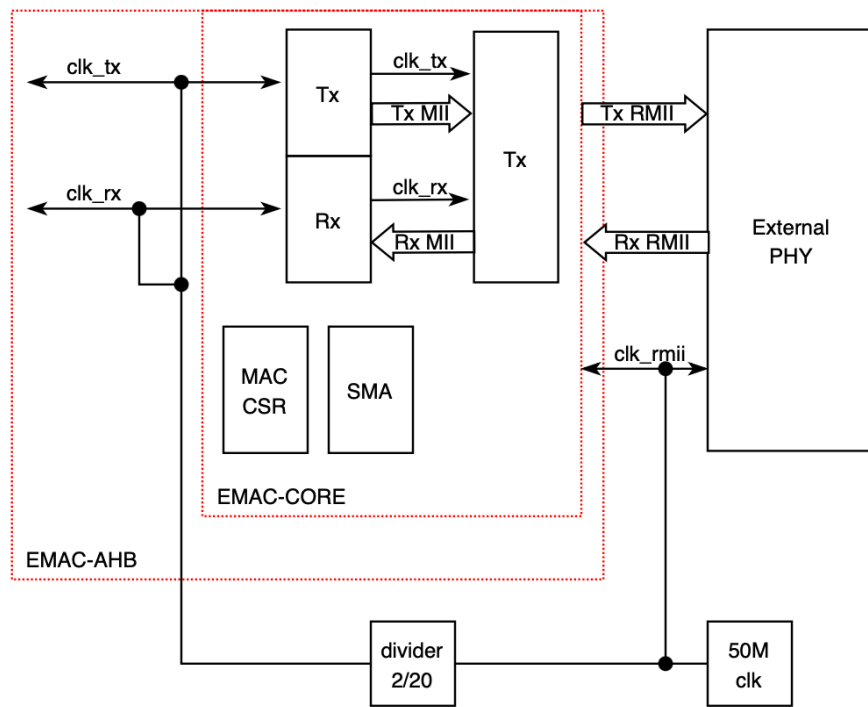


Figure 52.4-6. RMII Clock

52.4.4.3 Station Management Agent (SMA) Interface

As Figure 52.4-3 and Figure 52.4-5 show, the MAC uses MDC and MDIO signals to transfer control and data information to the PHY. The MDC clock is generated from the application clock through a clock divider and operates at a maximum frequency of 2.5 MHz. The MDIO data signal transfers to and from PHY synchronously with the MDC clock signal. Meanwhile, the PHY transmits register data.

- Read a PHY register:
 - Read [MIIBUSY](#) until the MII is in an idle state.
 - Configure [MIIDEV](#) to specify the PHY device to access.
 - Configure [MIIREG](#) to specify the address of the PHY register to access.
 - Configure [MIICSRCLK](#) to set the APB clock frequency.
 - Set [MIIWRITE](#) to 0 for read access.
 - Read [MIIBUSY](#) until the MII is in an idle state.
 - Read [EMACMIIDATA_REG](#) to retrieve the value of the PHY register.
- Write a PHY register:
 - Read [MIIBUSY](#) until the MII is in an idle state.
 - Configure [EMACMIIDATA_REG](#) to set the value to be written to the PHY register.
 - Configure [MIIDEV](#) to specify the PHY device to access.
 - Configure [MIIREG](#) to specify the address of the PHY register to access.
 - Configure [MIICSRCLK](#) to set the APB clock frequency.

- Set [MIWRITE](#) to 1 for write access.
- Read [MIIBUSY](#) until the MII is in an idle state, which indicates the completion of the write operation.

52.4.5 MAC Address Filtering

Address filtering checks the destination and source addresses on all received frames and reports the address filtering status accordingly. Address checking is based on different parameters (Frame filter register) chosen by the application. The filtered frame can also be identified: multicast or broadcast frame. Address filtering uses the station's physical (MAC) address for address checking.

52.4.5.1 Unicast Destination Address Filtering

The MAC supports up to 9 MAC addresses for unicast perfect filtering. MAC compares all 48 bits of the received unicast address with the programmed MAC address for any match. By default, [EMACADDR0](#) is always enabled, and the other addresses ([EMACADDR0](#) ~ [EMACADDR8](#)) are selected with an individual enable bit. Each byte of these other addresses ([EMACADDR0](#) ~ [EMACADDR8](#)) can be masked during comparison with the corresponding received DA byte by setting the corresponding Mask Byte Control bit in the register. This helps group address filtering for the DA.

52.4.5.2 Multicast Destination Address Filtering

The MAC can be programmed to pass all multicast frames by setting the Pass All Multicast ([PAM](#)) bit in the Frame Filter Register to 1. If the [PAM](#) bit is reset, the MAC the filtering for multicast addresses.

In perfect filtering mode, the multicast address is compared with the programmed MAC Destination Address Registers ([EMACADDR0](#) ~ [EMACADDR8](#)). Group address filtering is also supported.

52.4.5.3 Broadcast Address Filter

The MAC does not filter any broadcast frames in the default mode. However, if the MAC is programmed to reject all broadcast frames by setting the Disable Broadcast Frames ([DBF](#)) bit in the Frame Filter Register to 1, all broadcast frames will be dropped.

52.4.5.4 Unicast Source Address Filter

The MAC can also perform perfect filtering based on the source address field of the received frames. By default, the Address Filtering Module (AFM) compares the Source Address (SA) field with the values programmed in the SA register. The MAC address registers [1:3] can be configured to contain SA instead of DA for comparison, by setting bit[30] in the corresponding register. Group filtering with SA is also supported. The frames that fail the SA filter are dropped by the MAC if the Source Address Filter Enable ([SAFE](#)) bit in the Frame filter register is set. Otherwise, the result of the SA filter is given as a status bit in the Receive Status word (see Table 52.4-7).

When the [SAFE](#) bit is set to 1, the result of the SA filtering and DA filtering is AND'ed to decide whether the frame needs to be forwarded. This means that either of the filter failure results will drop the frame. Both filters have to pass the frame for the frame to be forwarded to the application.

52.4.5.5 Inverse Filtering Operation

For both destination and source address filtering, there is an option to invert the filter-match result at the final output. These are controlled by the **DAIF** and **SAIF** bits in the Frame filter register, respectively. The **DAIF** bit is applicable for both Unicast and Multicast DA frames. When the **DAIF** bit is set, the result of the unicast/multicast destination address filter is inverted. Similarly, when the **SAIF** bit is set, the result of the unicast SA filter is inverted.

The following two tables summarize the destination address and source address filtering based on the type of the frames received.

Table 52.4-14. Destination Address Filtering

Frame Type	PM	PF	DAIF	PAM	DB	DA Filter Operation
Broadcast	1	X	X	X	X	Pass
	0	X	X	X	0	Pass
	0	X	X	X	1	Fail
Unicast	1	X	X	X	X	Pass all frames
	0	X	0	X	X	Pass on perfect/group filter match
	0	X	1	X	X	Fail on perfect/group filter match
	0	1	0	X	X	Pass on perfect/group filter match
	0	1	1	X	X	Fail on perfect/group filter match
Multicast	1	X	X	X	X	Pass all frames
	X	X	X	1	X	Pass all frames
	0	X	0	0	X	Pass on perfect/group filter match and drop pause control frames if PCF = 0x.
	0	1	0	0	X	Pass on perfect/group filter match and drop pause control frames if PCF = 0x.
	0	X	1	0	X	Fail on perfect/group filter match and drop pause control frames if PCF = 0x.
	0	1	1	0	X	Fail on perfect/group filter match and drop pause control frames if PCF = 0x.

The filtering parameters in the MAC Frame Filter Register described in Table 52.4-14 are as follows.

Parameter name	Parameter setting
PM: Pass All Multicast	1: Set
PF: Perfect Filter	0: Cleared
DAIF: Destination Address Inverse Filtering	X: Don't care
PAM: Pass All Multicast	
DB: Disable Broadcast Frames	

Table 52.4-15. Source Address Filtering

Frame Type	PM	SAIF	SAF	Source Address Filter Operation
Unicast	1	X	X	Pass all frames
	0	0	0	Pass on perfect/group filter match but do not drop frames that fail
	0	0	0	Fail on perfect/group filter match but do not drop frames that fail
	0	0	0	Pass on perfect/group filter match and drop frames that fail
	0	0	0	Fail on perfect/group filter match and drop frames that fail

The filtering parameters in the MAC Frame Filter Register described in Table 52.4-15 are as follows.

Parameter name	Parameter setting
PM: Pass All Multicast	1: Set
SAF: Source Address Filtering	0: Cleared
SAIF: Source Address Inverse Filtering	X: Don't care

52.4.6 Energy Efficient Ethernet (EEE)

Energy Efficient Ethernet (EEE) is an optional operating mode that enables the MAC sublayer along with a series of physical layers to operate in the Low Power Idle (LPI) mode. The EEE mode supports 100 Mbps MAC operations compliant with the IEEE 802.3az-2010 standard. The EEE is supported only in the full-duplex mode when MAC uses MII.

The LPI mode allows power saving by switching off parts of the communication device functionality when there is no data to be transmitted and received. The system on both sides of the link can disable some functionalities and save power during the period of low-link utilization. The MAC controls whether the system should enter or exit the LPI mode and communicate this to the PHY.

52.4.6.1 Transmission

To enter the LPI mode, the software must set the [LPIEN](#) bit to indicate to the MAC to stop transmission and initiate the LPI protocol. The MAC completes the transmission in progress, generates its transmission status, and then starts transmitting the LPI pattern instead of the IDLE pattern during the Interframe gap (IFG). The MAC then updates [TLPIEN](#) and generates an interrupt.

To exit the LPI mode, the software must clear the [LPIEN](#) bit to indicate to the MAC. The MAC stops transmitting the LPI pattern, and starts transmitting the IDLE pattern. When the [LPI_TW_TIMER](#) expires, the MAC updates [TLPIEX](#) and generates an interrupt.

52.4.6.2 Reception

When the PHY receives the signals from the link partner to enter into the LPI state, the PHY starts transmitting the LPI pattern. The MAC updates [RLPIEN](#) and generates an interrupt.

When the PHY receives signals from the link partner to exit the LPI state, the PHY stops transmitting the LPI pattern. The MAC updates [RLPIEX](#) and generates an interrupt.

52.4.7 Source Address, VLAN, and CRC Control

52.4.7.1 Source Address Control

The software can enable source address insertion or replacement for all or some frames:

- Enable source address insertion or replacement for all frames by configuring [SAIRC](#).
- Enable source address insertion or replacement for the current frame by configuring [TDES1\[31:29\]](#) of the frame. When [TDES1\[31\]](#) is set, [MAC_ADDRESS1_HI](#) and [EMACADDR1LOW_REG](#) are used for insertion and replacement. When [TDES1\[31\]](#) is cleared, [MAC_ADDRESS0_HI](#) and [EMACADDR0LOW_REG](#) are used for insertion and replacement. When [ADDRESS_ENABLE1](#) is not set, [MAC_ADDRESS0_HI](#) and [EMACADDR0LOW_REG](#) are used for insertion and replacement irrespective of [TDES1\[31\]](#).

52.4.7.2 VLAN Control

The software can enable VLAN insertion, replacement, or deletion for all or some frames:

- Enable VLAN insertion, replacement, or deletion for all frames by configuring [VLC](#).
- Enable VLAN insertion, replacement, or deletion for the current frame by configuring [TDES0\[19:18\]](#) of the frame.

52.4.7.3 CRC Control

CRC replacement is a feature that the software can use to indicate the MAC to replace the FCS field of the frame being transmitted with the calculated CRC. This feature can be enabled for the current frame by configuring [TDES0\[24\]](#).

CRC replacement is valid only when MAC does not append CRC ([TDES0\[27\]](#) is set), i.e., when FCS is provided by software in frames transmitted. If source address or VLAN control is enabled, the MAC will replace or append FCS fields with CRC when [TDES0\[27\]](#) is set or cleared respectively.

52.4.8 Time Stamp

The IEEE 1588-2008 standard defines the Precision Time Protocol (PTP) that allows precise clock synchronization in measurement and control systems implemented with technologies such as local computing, network communication and distributed objects. It supports system-wide synchronization accuracy in the sub-microsecond range with a minimum network and local clock computing resources.

52.4.8.1 Transmit Path

The MAC captures a timestamp when the start-of-frame data (SFD) is sent on the MII interface. The frames for which timestamps are captured can be controlled on a per-frame basis. In other words, each transmit frame can be marked to indicate whether a timestamp should be captured for that frame. The MAC does not process the transmitted to identify PTP frames. Rather, it captures the timestamp or not according to the configurations of [TDES0\[25\]](#) by software. The MAC returns the timestamp to the software inside the corresponding transmit descriptor, thus connecting the timestamp automatically to the specific PTP frame. The 64-bit timestamp information is written to the [TDES6](#) and [TDES7](#) fields.

52.4.8.2 Receive Path

The MAC processes the received frames to identify valid PTP frames. The snapshot of the time to be sent to the application can be controlled by software via [EMACTSTPCTRL_REG](#).

The MAC returns the timestamp to the software inside the corresponding receive descriptor. The extended status (including the timestamp status and IPC status) is written to [RDES4](#), and the timestamp snapshot is written to [RDES6](#) and [RDES7](#).

52.4.9 Remote Wakeup

The MAC supports remote wakeup. When the receiver receives a remote wakeup frame or a magic packet, an interrupt will be triggered, and the PMU will be notified to wake up the SoC system.

The software can control whether to use remote wakeup frames to trigger interrupts by configuring [RWKPKTEN](#), whether to use magic packets to trigger interrupts by configuring [MGKPKTEN](#), and whether the MAC drops all received frames in power-down mode by configuring [PWRDWN](#).

52.4.10 Good Transmitted and Received Frames

A frame that has been successfully transmitted is considered a “good frame”. In other words, a transmitted frame is considered to be “good”, if the frame transmission is not aborted due to the following errors:

- Jabber timeout ([TDES0\[14\]](#))
- No carrier or loss of carrier ([TDES0\[11:10\]](#))
- Late collision ([TDES0\[9\]](#))
- Frame underflow ([TDES0\[1\]](#))
- Excessive deferral ([TDES0\[2\]](#))
- Excessive collision ([TDES0\[8\]](#))

A received frame is considered to be “good” if the following errors do not occur:

- CRC error ([RDES0\[1\]](#))
- Runt frames (frames shorter than 64 bytes)
- Alignment error (in 10/100 Mbps modes only) ([RDES0\[2\]](#))
- Length error (non-type frames only) ([RDES0\[12\]](#))
- Frame size over the maximum size (for non-type frames over the maximum frame size only) ([RDES0\[7\]](#))
- MII_RXER input error ([RDES0\[3\]](#))

The maximum frame size depends on the frame type:

- The maximum size of untagged frames = 1518 bytes
- The maximum size of VLAN frames = 1522 bytes

52.5 Programming Procedures

52.5.1 MAC System Layer Configuration

1. Configure the IO interfaces. For detailed configurations about the MII and the RMII interface, see Chapter [9 GPIO Matrix and IO MUX](#).

2. Configure the EMAC system clock by setting [HP_SYS_CLKRST_EMAC_SYS_CLK_EN](#) to 1.
3. Configure the PHY interface clock:
 - RMII interface: select the RMII interface by configuring [HP_SYS_PHY_INTF_SEL](#) to 4.
 - Select RMII reference clock source:
 - * Reference clock sourced from the external crystal:
 - Configure [HP_SYS_CLKRST_PAD_EMAC_REF_CLK_EN](#) to disable [PAD_EMAC_REF_CLK](#) outputted to GPIO.
 - Configure [HP_SYS_CLKRST_REF_50M_CLK_EN](#) to disable [PLL_F50M_CLK](#).
 - * Reference clock sourced from ESP32-P4:
 - Configure [HP_SYS_CLKRST_PAD_EMAC_REF_CLK_EN](#) to enable [PAD_EMAC_REF_CLK](#) outputted to GPIO.
 - Configure [HP_SYS_CLKRST_REF_50M_CLK_DIV_NUM](#) to specify the clock divisor for [PLL_F50M_CLK](#).
 - Configure [HP_SYS_CLKRST_REF_50M_CLK_EN](#) to enable [PLL_F50M_CLK](#).
 - Configure [HP_SYS_CLKRST_EMAC_RMII_CLK_SRC_SEL](#) to specify the clock source of [EMAC_RMII_CLK](#) as [PAD_EMAC_TXRX_CLK](#);
 - Configure [HP_SYS_CLKRST_EMAC_RX_CLK_SRC_SEL](#) to specify the clock source of [EMAC_RX_CLK](#) as [PAD_EMAC_TXRX_CLK](#);
 - Configure [HP_SYS_CLKRST_EMAC_RX_CLK_DIV_NUM](#) to specify the clock divisor for [EMAC_RX_CLK](#) according to the EMAC line speed (10 Mbit/s or 100 Mbit/s);
 - Configure [HP_SYS_CLKRST_EMAC_TX_CLK_SRC_SEL](#) to specify the clock source of [EMAC_TX_CLK](#) as [PAD_EMAC_TXRX_CLK](#);
 - Configure [HP_SYS_CLKRST_EMAC_TX_CLK_DIV_NUM](#) to specify the clock divisor for [EMAC_TX_CLK](#) according to the EMAC line speed (10 Mbit/s or 100 Mbit/s);
 - Configure [LP_AONCLKRST_HP_PAD_EMAC_TXRX_CLK_EN](#) to enable the clock from IO pin [PAD_EMAC_TXRX_CLK](#);
 - Configure [HP_SYS_CLKRST_EMAC_RMII_CLK_EN](#) to enable [EMAC_RMII_CLK](#);
 - Configure [HP_SYS_CLKRST_EMAC_RX_CLK_EN](#) to enable [EMAC_RX_CLK](#);
 - Configure [HP_SYS_CLKRST_EMAC_TX_CLK_EN](#) to enable [EMAC_TX_CLK](#);
 - MII interface: select the MII interface by configuring [HP_SYS_PHY_INTF_SEL](#) to 0.
 - Configure [HP_SYS_CLKRST_EMAC_RX_CLK_SRC_SEL](#) to specify [EMAC_RX_CLK](#) as [PAD_EMAC_RX_CLK](#);
 - Configure [HP_SYS_CLKRST_EMAC_RX_CLK_DIV_NUM](#) to specify the clock divisor for [EMAC_RX_CLK](#) according to the EMAC line speed (10 Mbit/s or 100 Mbit/s);
 - Configure [HP_SYS_CLKRST_EMAC_TX_CLK_SRC_SEL](#) to specify [EMAC_TX_CLK](#) as [PAD_EMAC_TX_CLK](#);

- Configure [HP_SYS_CLKRST_EMAC_TX_CLK_DIV_NUM](#) to specify the clock divisor for EMAC_TX_CLK according to the EMAC line speed (10 Mbit/s or 100 Mbit/s);
- Configure [LP_AONCLKRST_HP_PAD_EMAC_RX_CLK_EN](#) to enable the clock from IO pin PAD_EMAC_RX_CLK;
- Configure [LP_AONCLKRST_HP_PAD_EMAC_TX_CLK_EN](#) to enable the clock from IO pin PAD_EMAC_TX_CLK;
- Configure [HP_SYS_CLKRST_EMAC_RX_CLK_EN](#) to enable EMAC_RX_CLK;
- Configure [HP_SYS_CLKRST_EMAC_TX_CLK_EN](#) to enable EMAC_TX_CLK;

52.5.2 EMAC Initial Configuration

Initialize EMAC_DMA:

- Reset EMAC by setting [SW_RST](#) to 1.
- Wait for the completion of reset until reading 0 from [SW_RST](#), which indicates that the reset is done.
- Read [AHB_ST](#); a value of 0 indicates that all AHB bus transactions have been completed.
- Configure the AHB bus mode for EMAC_DMA via [DMABUSMODE_REG](#).
- Prepare the linked list and configure the linked list base address via [DMARXBASEADDR_REG](#) and [DMATXBASEADDR_REG](#).
- Configure the operating mode for EMAC_DMA and EMAC_MTL via [DMAOPERATION_MODE_REG](#).
- Configure whether to enable DMA interrupts via [DMAIN_EN_REG](#).

Initialize EMAC_CORE:

- Read [EMACMIIDADDR_REG](#) and [EMACMIIDATA_REG](#) for the PHY's connection status, operating frequency, and operating mode.
- Configure [EMACADDROHIGH_REG](#) and [EMACADDROLOW_REG](#) to specify the MAC address.
- Configure [EMACFF_REG](#) to select the packet filtering mode.
- Configure [EMACFC_REG](#) for flow control.
- Configure whether to mask MAC interrupts via [EMACINTMASK_REG](#).
- Configure the MAC's operating mode for transmission and reception via [EMACCONFIG_REG](#).

52.5.3 Starting Transmission

- Start EMAC_DMA transmission by setting [START_STOP_TRANSMISSION_COMMAND](#) to 1.
- Start EMAC_CORE transmission by setting [EMACTX](#) to 1.
- Detect the TRANS_INT interrupt if enabled, and wait for the frame transmission to complete.

52.5.4 Starting Reception

- Start EMAC_DMA reception by setting [START_STOP_RX](#) to 1.

- Start EMAC_CORE reception by setting [EMACRX](#) to 1.
- Detect the RECV_INT interrupt if enabled, and wait for the frame reception to complete.

52.5.5 TX Entering and Exiting the LPI State

- Configure the LPI timer counter via [EMACLPITIMERSCONTROL_REG](#)
- Make the transmitter to enter the LPI state via [EMACLPI_CSR_REG](#)
- Detect the LPI_INT interrupt, and wait for [TLPIEN](#) to be set, which indicates that the transmitter has entered the LPI state
- Bring the transmitter out of the LPI state by clearing [LPIEN](#)
- Detect the LPI_INT interrupt, and wait for [TLPIEX](#) to be set, which indicates that the transmitter has exited the LPI state

52.5.6 RX Entering and Exiting the LPI State

- Detect the LPI_INT interrupt, and wait for [RLPIEN](#) to be set, which indicates that the receiver has entered the LPI state
- Detect the LPI_INT interrupt, and wait for [RLPIEX](#) to be set, which indicates that the receiver has exited the LPI state

52.6 Interrupts

ESP32-P4's EMAC can generate the following interrupt signals that will be sent to the [Interrupt Matrix](#).

- ETH_MAC_INTR
- PMT_INTR
- LPT_INTR

There are several internal interrupt sources from EMAC that can generate the above interrupt signals.

The following interrupt sources can generate the ETH_MAC_INTR interrupt signal:

- [ABN_INT_SUMM](#): Triggered when [DMAIN_AISE](#) is 1 and the following interrupts are triggered.
 - [TRANS_PROC_STOP](#): Triggered when [DMAIN_TSE](#) is 1 and the transmit process stops.
 - [TRANS_JABBER_TO](#): Triggered when [DMAIN_TJTE](#) is 1 and the frame size exceeds 2048 bytes (10240 bytes if the jumbo frame is enabled).
 - [RECV_OVERFLOW](#): Triggered when [DMAIN_OIE](#) is 1 and the receive buffer overflows during frame reception.
 - [TRANS_UNDFLOW](#): Triggered when [DMAIN_UIE](#) is 1 and the transmit buffer underflows during frame transmission.
 - [RECV_BUF_UNAVAIL](#): Triggered when [DMAIN_RBUE](#) is 1, the Host owns the Next Descriptor in the Receive List and the DMA cannot acquire it.
 - [RECV_PROC_STOP](#): Triggered when [DMAIN_RSE](#) is 1 and the receive process stops.

- [RECV_WDT_TO](#): Triggered when [DMAIN_RWTE](#) is 1 and the Receive Watchdog Timer has expired while receiving the current frame.
- [EARLY_TRANS_INT](#): Triggered when [DMAIN_ETIE](#) is 1 and the frame has been completely transmitted to MTL TX FIFO.
- [FATAL_BUS_ERR_INT](#): Triggered when [DMAIN_FBEE](#) is 1 and a bus error described in [ERROR_BITS](#) occurs.
- [NORM_INT_SUMM](#): Triggered when [DMAIN_NISE](#) is 1 and the following interrupts are triggered.
 - [TRANS_INT](#): Triggered when [DMAIN_TIE](#) is 1 and the transmission completes.
 - [TRANS_BUF_UNAVAIL](#): Triggered when [DMAIN_TBUE](#) is 1, the Host owns the Next Descriptor in the Transmit List and the DMA cannot acquire it.
 - [RECV_INT](#): Triggered when [DMAIN_RIE](#) is 1 and the frame has been received.
 - [EARLY_RECV_INT](#): Triggered when [DMAIN_ERIE](#) is 1 and the DMA has received the first buffer of the data packet.

[EMAC_PMT_INT](#): Triggered when [PMTINTMASK](#) is 0, and a magic packet or a remote wakeup frame has been received in power-down mode.

[TS_TRI_INT](#): Triggered when [TSINTMASK](#) is 0 and any of the following conditions is true:

- The system time value equals or exceeds the value specified in the Target Time registers
- There is an overflow in the System Time Seconds Register

[EMAC_LPI_INT](#): Triggered when [LPIINTMASK](#) is 0 and any of the following conditions is true:

- The MAC Transmitter enters the LPI state
- The MAC Transmitter exits the LPI state
- The MAC Receiver enters the LPI state
- The MAC Receiver exits the LPI state

The following interrupt sources can generate the [PMT_INTR](#) interrupt signal:

- [RWKPRCVD](#): Triggered when a remote wakeup frame is received.
- [MGKPRCVD](#): Triggered when a magic packet is received.

The following interrupt source can generate the [LPI_INTR](#) interrupt signal:

- [RLPIEX](#): Triggered when the MAC receiver has stopped receiving the LPI pattern on PHY and has exited the LPI state.

52.7 Register Summary

The addresses in this section are relative to EMAC base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
DMA configuration and control registers			
DMABUSMODE_REG	Bus mode register	0x1000	R/WS/SC
DMATXPOLLDEMAND_REG	Transmit poll demand register	0x1004	RO/WT
DMARXPOLLDEMAND_REG	Receive poll demand register	0x1008	RO/WT
DMARXBASEADDR_REG	Base address of the first receive descriptor	0x100C	R/W
DMATXBASEADDR_REG	Base address of the first transmit descriptor	0x1010	R/W
DMASTATUS_REG	Base address of interrupt, error and other events	0x1014	R/SS/WC
DMAOPERATION_MODE_REG	Operation mode and command register	0x1018	R/SS/WC
DMAIN_EN_REG	Interrupt disable and enable register	0x101C	R/W
DMAMISSEDFR_REG	Missed frame and buffer overflow counter register	0x1020	R/W
DMARINTWDTIMER_REG	Receive watchdog timer counter register	0x1024	R/W
DMAAHBSTATUS_REG	AHB master interface status register	0x102C	RO
DMATXCURRDESC_REG	Current transmit descriptor register	0x1048	RO
DMARXCURRDESC_REG	Current receive descriptor register	0x104C	RO
DMATXCURRADDR_BUF_REG	Current transmit buffer address register	0x1050	RO
DMARXCURRADDR_BUF_REG	Current receive buffer address register	0x1054	RO
MAC configuration and control registers			
EMACCONFIG_REG	MAC configuration register	0x0000	R/W
EMACFF_REG	Frame filter register	0x0004	R/W
EMACMIIADDR_REG	PHY access permission configuration register	0x0010	R/WS/SC
EMACMIIDATA_REG	PHY write and read data register	0x0014	R/W
EMACFC_REG	Frame flow control register	0x0018	varies
EMACVLANTAG_REG	802.1Q VLAN tag register	0x001C	R/W
EMACDEBUG_REG	Status and debug register	0x0024	RO
PMT_RWUFR_REG	Wake-up frame filter register	0x0028	RO

Name	Description	Address	Access
PMT_CSR_REG	Power management control and status register	0x002C	RO
EMACLPI_CSR_REG	LPI control and status register	0x0030	RO
EMACLPITIMERSCONTROL_REG	LPI timer control register	0x0034	RO
EMACINTS_REG	Interrupt status register	0x0038	RO
EMACINTMASK_REG	Interrupt mask register	0x003C	R/W
EMACADDR0HIGH_REG	The higher 16 bits of the first 6-byte MAC address	0x0040	R/W
EMACADDR0LOW_REG	The lower 32 bits of the first 6-byte MAC address	0x0044	R/W
EMACADDR1HIGH_REG	The higher 16 bits of the second 6-byte MAC address	0x0048	R/W
EMACADDR1LOW_REG	The lower 32 bits of the second 6-byte MAC address	0x004C	R/W
EMACADDR2HIGH_REG	The higher 16 bits of the third 6-byte MAC address	0x0050	R/W
EMACADDR2LOW_REG	The lower 32 bits of the third 6-byte MAC address	0x0054	R/W
EMACADDR3HIGH_REG	The higher 16 bits of the fourth 6-byte MAC address	0x0058	R/W
EMACADDR3LOW_REG	The lower 32 bits of the fourth 6-byte MAC address	0x005C	R/W
EMACADDR4HIGH_REG	The higher 16 bits of the fifth 6-byte MAC address	0x0060	R/W
EMACADDR4LOW_REG	The lower 32 bits of the fifth 6-byte MAC address	0x0064	R/W
EMACADDR5HIGH_REG	The higher 16 bits of the sixth 6-byte MAC address	0x0068	R/W
EMACADDR5LOW_REG	The lower 32 bits of the sixth 6-byte MAC address	0x006C	R/W
EMACADDR6HIGH_REG	The higher 16 bits of the seventh 6-byte MAC address	0x0070	R/W
EMACADDR6LOW_REG	The lower 32 bits of the seventh 6-byte MAC address	0x0074	R/W
EMACADDR7HIGH_REG	The higher 16 bits of the eighth 6-byte MAC address	0x0078	R/W
EMACADDR7LOW_REG	The lower 32 bits of the eighth 6-byte MAC address	0x007C	R/W
EMACADDR8HIGH_REG	The higher 16 bits of the ninth 6-byte MAC address	0x0080	R/W
EMACADDR8LOW_REG	The lower 32 bits of the ninth 6-byte MAC address	0x0084	R/W
EMACCSTATUS_REG	Interconnection status	0x00D8	RO

Name	Description	Address	Access
EMACWDOGTO_REG	Watchdog timer timeout control register	0x00DC	R/W
EMACTXVLTCTRL_REG	TX VLAN tag control register	0x0584	R/W
EMACTSTPCTRL_REG	Timestamp control register	0x0700	R/W
EMACSUB2NDINCR_REG	Sub-second increment register	0x0704	R/W
EMACSYSTIM2ND_REG	System time second register	0x0708	RO
EMACSYSTIMNS_REG	System time nanosecond register	0x070C	RO
EMAC2NDUPT_REG	System time second update register	0x0710	R/W
EMACNSUPT_REG	System time nanosecond update register	0x0714	R/W
EMACTSTPADDEND_REG	Fine update mode addend	0x0718	R/W
EMACTGTIME2ND_REG	Target time second register	0x071C	R/W
EMACTGTIMENS_REG	Target time nanosecond register	0x0720	R/W
EMACSYSTIMHIGHWORD2ND_REG	System time nanosecond register (bit 32 to bit 47)	0x0724	R/W/SU
EMACTSTPSTATUS_REG	Timestamp status register	0x0728	R/SS/RC

52.8 Registers

The addresses in this section are relative to EMAC base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 52.1. DMABUSMODE_REG (0x1000)

(DMAREBUILDINCRBURST)				(reserved)				DMATransmitPriority				DMAAddralibea				RX_DMA_PBL				FIXED_BURST				PRI_RATIO				PROG_BURST_LEN				ALT_DESC_SIZE				DESC_SKIP_LEN				DMA_ARB_SCH			
SW_RST																																											
31	30	28	27	26	25	24	23	22	17	16	15	14	13	8	7	6	2	1	0																								
0	0	0	0	0	0	0	0	0	0x01	0	0x0	0x01	0	0	0x00	0	0	1	Reset																								

DMAREBUILDINCRBURST Configures whether to rebuild INCRx burst.

0: Not rebuild

1: If the AHB master gets an EBT (retry, split, or losing bus grant), the AHB master interface rebuilds the pending beats of any burst transfer initiated with INCRx using INCRx and SINGLE.

(R/W)

DMATransmitPriority Configures whether the transmit DMA has higher priority than the receive DMA during arbitration.

0: Lower priority

1: Higher priority

(R/W)

DMAMixedBurst Configures whether to use mixed burst.

0: Not use

1: If the FIXED_BURST bit is 1, the AHB Master interface starts all bursts of length more than 16 with INCR (undefined burst) whereas it reverts to fixed burst transfers (INCRx and SINGLE) for burst length of 16 and less.

(R/W)

DMAAddralibea Configures whether bursts are aligned to address.

0: Not aligned to address

1: If the FIXED_BURST bit is 1, the AHB interface generates all bursts aligned to the start address LS bits. If the FIXED_BURST bit is 0, the first burst (accessing the data buffer's start address) is not aligned, but subsequent bursts are aligned to the address.

(R/W)

Continued on the next page...

Register 52.1. DMABUSMODE_REG (0x1000)

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PBLX8_MODE Configures whether to multiply the PBL (programmable burst length) value by 8.

0: Disable

1: The programmed PBL value (PROG_BURST_LEN and RX_DMA_PBL) is multiplied by 8. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value.

(R/W)

USE_SEP_PBL Configures whether to use separate PBL.

0: PROG_BURST_LEN is applicable only for TX DMA, and RX DMA uses RX_DMA_PBL.

1: PROG_BURST_LEN is applicable for both TX and RX DMA.

(R/W)

RX_DMA_PBL Configure the maximum number of beats to be transferred in one RX DMA transaction, namely the maximum value that is used in a single block Read or Write.

Value range: 1, 2, 4, 8, 16 and 32. Any other value results in undefined behavior.

Valid only when USE_SEP_PBL is set.

(R/W)

FIXED_BURST Configures whether the AHB Master interface performs fixed burst transfers.

0: The AHB interface uses SINGLE and INCR burst transfer operations.

1: The AHB interface uses only SINGLE, INCR4, INCR8, or INCR16 during start of the normal burst transfers.

(R/W)

PRI_RATIO Configures the priority ratio in the weighted round-robin arbitration between the RX DMA and TX DMA.

0: 1:1

1: 2:0

2: 3:1

3: 4:1

Valid only when DMA_ARB_SCH is 0. (R/W)

PROG_BURST_LEN Configures the maximum number of beats to be transferred in one DMA transaction.

If the number of beats to be transferred is more than 32, then perform the following steps:

1. Set the PBLx8 mode with PBLX8_MODE

2. Set the PBL with PROG_BURST_LEN

(R/W)

ALT_DESC_SIZE Configures whether to increase the descriptor size to 32 bytes.

0: Keep 16 bytes

1: Increase to 32 bytes

(R/W)

DESC_SKIP_LEN Configures the number of Word, Dword, or Lword (depending on the 32-bit, 64-bit, or 128-bit bus) to skip between two unchained descriptors.

The address skipping starts from the end of the current descriptor to the start of next descriptor. When this field is 0, the descriptor table is taken as contiguous by the DMA in Ring mode. (R/W)

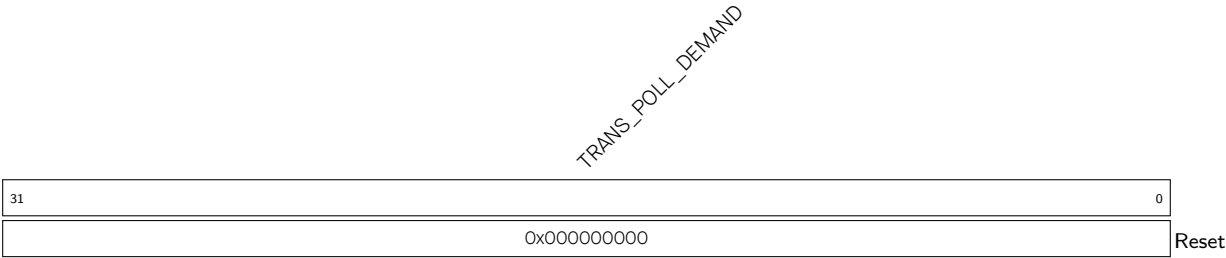
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Register 52.1. DMABUSMODE_REG (0x1000)

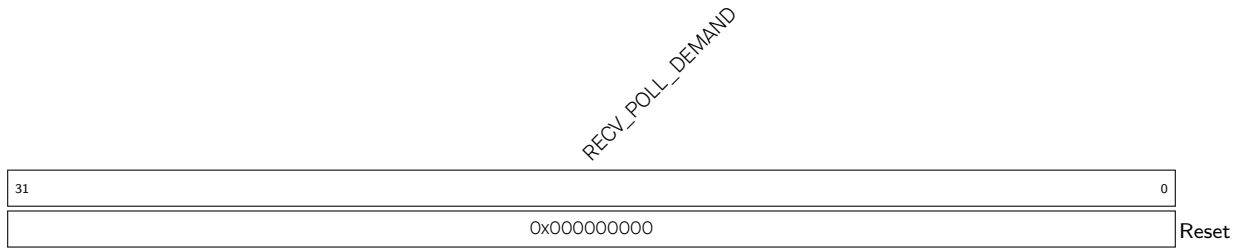
Continued from the previous page...

- DMA_ARB_SCH** Configures the arbitration scheme between the transmit and receive paths.
0: Weighted round-robin with RX:TX or TX:RX. In this case, the priority between the paths is according to the priority specified in PRI_RATIO.
1: Fixed priority (The transmit path has priority over receive path).
(R/W)
- SW_RST** Configures whether the MAC DMA Controller resets the logic and all internal registers of the MAC.
0: Release from reset
1: Reset
This bit is cleared automatically after the reset operation has completed in all of the ETH_MAC clock domains.
(R/WS/SC)

Register 52.2. DMATXPOLLDEMAND_REG (0x1004)



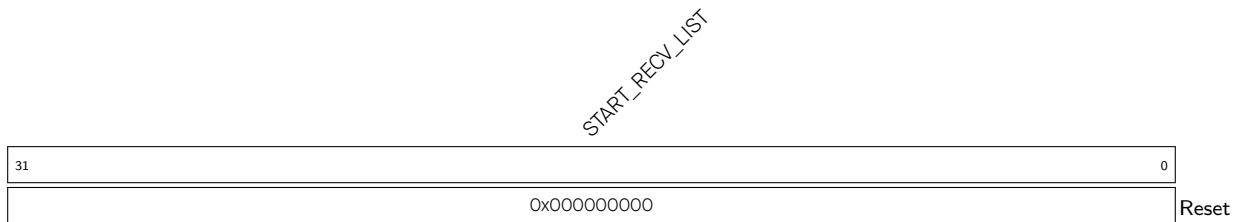
- TRANS_POLL_DEMAND** Configures whether to enable the TX DMA to check if the DMA owns the current descriptor.
Any value: Enable
When this field is written, the DMA reads the current descriptor pointed to by [DMATXCUR-RDESC_REG](#). If that descriptor is not available (owned by the Host), the transmission returns to the Suspend state and the Bit 2 (TU) of Register 5 (Status Register) is asserted. If the descriptor is available, the transmission resumes. (RO/WT)

Register 52.3. DMARXPOLLDEMAND_REG (0x1008)

RECV_POLL_DEMAND Configures whether to enable the RX DMA to check if the DMA owns the current descriptor.

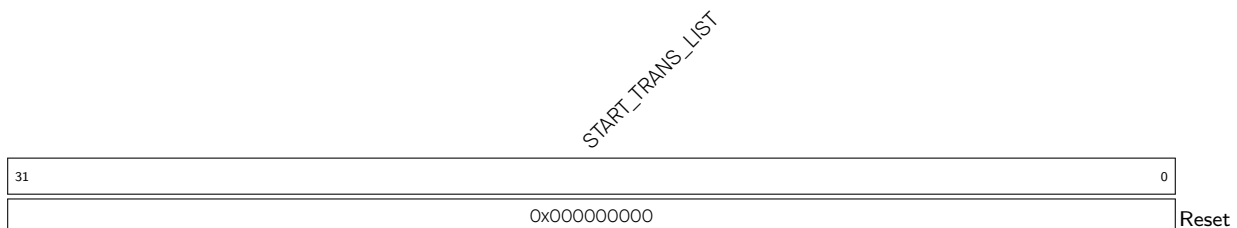
Any value: Enable

When this field is written, the DMA reads the current descriptor pointed to by [DMARXCUR-RDESC_REG](#). If that descriptor is not available (owned by the Host), the reception returns to the Suspended state and the Bit 7 (RU) of Register 5 (Status Register) is not asserted. If the descriptor is available, the RX DMA returns to the active state.(RO/WT)

Register 52.4. DMARXBASEADDR_REG (0x100C)

START_RECV_LIST Configures the base address of the first descriptor in the receive descriptor linked list.

The LSB bits (1:0) are ignored (32-bit wide bus) and internally taken as all-zero by the DMA. Therefore, these LSB bits are read-only (RO). (R/W)

Register 52.5. DMATXBASEADDR_REG (0x1010)

START_TRANS_LIST Configures the base address of the first descriptor in the Transmit Descriptor list. The LSB bits (1:0) are ignored (32-bit wide bus) and are internally taken as all-zero by the DMA. Therefore, these LSB bits are read-only (RO). (R/W)

Register 52.6. DMASTATUS_REG (0x1014)

(reserved)		EMAC_LPI_INT		TS_TRI_INT		EMAC_PMT_INT		(reserved)		ERROR_BITS		TRANS_PROC_STATE		RECV_PROC_STATE		NORM_INT_SUMM		ABN_INT_SUMM		EARLY_RECV_INT		FATAL_BUS_ERR_INT		(reserved)		EARLY_TRANS_INT		RECV_WDT_TO		RECV_PROC_STOP		RECV_BUF_UNAVAIL		TRANS_UNDFLOW		TRANS_OVFLOW		TRANS_JABBER_TO		TRANS_BUF_UNAVAIL		TRANS_PROC_STOP		TRANS_INT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
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Reset

EMAC_LPI_INT The raw interrupt status of EMAC_LPI_INT. (RO)

TS_TRI_INT The raw interrupt status of TS_TRI_INT. (RO)

EMAC_PMT_INT The raw interrupt status of EMAC_PMT_INT. (RO)

ERROR_BITS Represents the type of error that caused a Bus Error, for example, error response on the AHB interface.

0: Error during RX DMA Write Data Transfer

1: Error during TX DMA Read Data Transfer

2: Error during RX DMA Descriptor Write Access

3: Error during TX DMA Descriptor Write Access

4: Error during RX DMA Descriptor Read Access

5: Error during TX DMA Descriptor Read Access

Valid only when Bit 13 (FBI) is 1. This field does not generate an interrupt. (RO)

TRANS_PROC_STATE Represents the Transmit DMA FSM state.

0: Stopped; Reset or Stop Transmit Command issued

1: Running; Fetching Transmit Transfer Descriptor

2: Running; Waiting for status

3: Running; Reading Data from Host memory buffer and queuing it to transmit buffer (TX FIFO)

4: TIME_STAMP write state

5: Reserved for future use

6: Suspended; Transmit Descriptor Unavailable or Transmit Buffer Underflow

7: Running; Closing Transmit Descriptor

This field does not generate an interrupt. (RO)

Continued on the next page...

Register 52.6. DMASTATUS_REG (0x1014)

Continued from the previous page...

NORM_INT_SUMM Represents the logical OR of the following when the corresponding interrupt bits are enabled in Interrupt Enable Register.

Bit[0]: Transmit Interrupt

Bit[2]: Transmit Buffer Unavailable

Bit[6]: Receive Interrupt

Bit[14]: Early Receive Interrupt

Only unmasked interrupts affect this bit.

This is a sticky bit and must be cleared (by writing 1 to this bit) each time a corresponding bit, which causes this set to be set, is cleared. (R/SS/WC)

ABN_INT_SUMM Represents the logical OR of the following when the corresponding interrupt bits are enabled in Interrupt Enable Register.

Bit[1]: Transmit Process Stopped

Bit[3]: Transmit Jabber Timeout

Bit[4]: Receive FIFO Overflow

Bit[5]: Transmit Underflow

Bit[7]: Receive Buffer Unavailable

Bit[8]: Receive Process Stopped

Bit[9]: Receive Watchdog Timeout

Bit[10]: Early Transmit Interrupt

Bit[13]: Fatal Bus Error

Only unmasked interrupts affect this bit.

This is a sticky bit and must be cleared each time a corresponding bit, which causes this bit to be set, is cleared. (R/SS/WC)

EARLY_RECV_INT The raw interrupt status of EARLY_RECV_INT. (R/SS/WC)

FATAL_BUS_ERR_INT The raw interrupt status of FATAL_BUS_ERR_INT. (R/SS/WC)

EARLY_TRANS_INT The raw interrupt status of EARLY_TRANS_INT. (R/SS/WC)

RCV_WDT_TO Represents whether the Receive Watchdog Timer expired while receiving the current frame and the current frame is truncated after the watchdog timeout.

0: Not expired

1: Expired

(R/SS/WC)

RCV_PROC_STOP Represents whether the Receive Process enters the Stopped state.

0: Not stopped

1: Stopped

(R/SS/WC)

Continued on the next page...

Register 52.6. DMASTATUS_REG (0x1014)

Continued from the previous page...

RECV_BUF_UNAVAIL Represents whether the Receive Buffer is available.

0: Available

1: Unavailable

This bit is set only when the previous Receive Descriptor is owned by the DMA.

When set, the Host owns the Next Descriptor in the Receive List and the DMA cannot acquire it. The Receive Process is suspended.

To resume processing Receive descriptors, the Host should change the ownership of the descriptor and issue a Receive Poll Demand command. If no Receive Poll Demand is issued, the Receive Process resumes when the next recognized incoming frame is received. (R/SS/WC)

RECV_INT The raw interrupt status of RECV_INT. (R/SS/WC)

TRANS_UNDFLOW Represents whether the Transmit Buffer had an Underflow during frame transmission.

0: No underflow

1: Underflow

When this bit is 1, transmission is suspended and an Underflow Error TDES0[1] is set. (R/SS/WC)

RECV_OVFLOW Represents whether the Receive Buffer had an Overflow during frame reception.

0: No overflow

1: Overflow

If the partial frame is transferred to the application, the overflow status is set in RDES0[11]. (R/SS/WC)

TRANS_JABBER_TO Represents whether the Transmit Jabber Timer expired, which happens when the frame size exceeds 2,048 (10,240 bytes when the Jumbo frame is enabled).

0: Not expired

1: Expired

When the Jabber Timeout occurs, the transmission process is aborted and placed in the Stopped state. This causes the Transmit Jabber Timeout TDES0[14] flag to assert. (R/SS/WC)

TRANS_BUF_UNAVAIL Represents whether the Transmit Buffer is available.

0: Available

1: Unavailable

When this bit is 1, the Host owns the Next Descriptor in the Transmit List and the DMA cannot acquire it. Transmission is suspended. Bits[22:20] explain the Transmit Process state transitions.

To resume processing Transmit descriptors, the Host should change the ownership of the descriptor by setting TDES0[31] and then issue a Transmit Poll Demand command. (R/SS/WC)

TRANS_PROC_STOP Represents whether the transmission is stopped.

0: Not stopped

1: Stopped

(R/SS/WC)

TRANS_INT The raw interrupt status of TRANS_INT. (R/SS/WC)

Register 52.7. DMAOPERATION_MODE_REG (0x1018)

(reserved)					DIS_DROP_TCPIP_ERR_FRAM (reserved)					DIS_FLUSH_RECV_FRAMES (reserved)					FLUSH_TX_FIFO (reserved)					TX_THRESH_CTRL START_STOP_TRANSMISSION_COMMAND (reserved)					(reserved)					FWD_ERR_FRAME FWD_UNDER_GF DROP_GFRM RX_THRESH_CTRL OPT_SECOND_FRAME START_STOP_RX (reserved)				
31					27	26	25	24	23	21		20	19	17		16	14		13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset		

DIS_DROP_TCPIP_ERR_FRAM Configures whether to disable dropping of TCP/IP checksum error frames.

0: Enable

1: Disable

When FWD_ERR_FRAME is 0, this bit is ignored and all error frames are dropped. (R/W)

DIS_FLUSH_RECV_FRAMES Configures whether to disable flushing of received frames because of the unavailability of receive descriptors or buffers.

0: Enable

1: Disable

(R/W)

FLUSH_TX_FIFO Configures whether to flush TX FIFO to default values.

0: Not flush

1: Flush

(R/WS/SC)

TX_THRESH_CTRL Configures the threshold of TX FIFO. Transmission starts when the frame size within the TX FIFO is larger than the threshold. In addition, full frames with a length less than the threshold are also transmitted.

0: 64

1: 128

2: 192

3: 256

4: 40

5: 32

6: 24

7: 16

Transmission starts when the frame size within the TX FIFO is larger than the threshold. In addition, full frames with a length less than the threshold are also transmitted.

(R/W)

Continued on the next page...

Register 52.7. DMAOPERATION_MODE_REG (0x1018)

Continued from the previous page...

START_STOP_TRANSMISSION_COMMAND Configures whether to start or stop the transmission.

0: Stop

1: Start

(R/W)

FWD_ERR_FRAME Configures whether RX FIFO forward frames with error status (CRC error, collision error, giant frame, watchdog timeout, or overflow)

0: Drop

1: Forward

(R/W)

FWD_UNDER_GF Configures whether the RX FIFO forward undersized (frames with no Error and length less than 64 bytes) good frames including PAD and CRC.

0: Drop

1: Forward

(R/W)

DROP_GFRM Configures whether to the ETH_MAC drops the received giant frames in the RX FIFO.

0: Not drop

1: Drop

(R/W)

RX_THRESH_CTRL Configures the threshold of the RX FIFO.

0: 64

1: 32

2: 96

3: 128

Transfer (request) to DMA starts when the frame size within the RX FIFO is larger than the threshold. (R/W)

OPT_SECOND_FRAME Configures whether the DMA processes the second frame of the Transmit data even before the status for the first frame is obtained.

0: Not process

1: Process

(R/W)

START_STOP_RX Configures whether to start or stop RX DMA reception.

0: Stop reception after the transfer of the current frame

1: Start reception

(R/W)

Register 52.8. DMAIN_EN_REG (0x101C)

(reserved)																	DMAIN_NISE DMAIN_AISE DMAIN_ERIE DMAIN_FBEE (reserved) DMAIN_ETIE DMAIN_RWTE DMAIN_RSE DMAIN_RBUE DMAIN_RIE DMAIN_UIE DMAIN_OIE DMAIN_TJTE DMAIN_TBUE DMAIN_TSE DMAIN_TIE																	
31																17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0								

DMAIN_NISE Write 1 to enable and write 0 to disable the summary of the following normal interrupt enable bits:

Bit[0]: Transmit Interrupt

Bit[2]: Transmit Buffer Unavailable

Bit[6]: Receive Interrupt

Bit[14]: Early Receive Interrupt

(R/W)

DMAIN_AISE Write 1 to enable and write 0 to disable the summary of the following abnormal interrupt enable bits:

Bit[1]: Transmit Process Stopped

Bit[3]: Transmit Jabber Timeout

Bit[4]: Receive FIFO Overflow

Bit[5]: Transmit Underflow

Bit[7]: Receive Buffer Unavailable

Bit[8]: Receive Process Stopped

Bit[9]: Receive Watchdog Timeout

Bit[10]: Early Transmit Interrupt

Bit[13]: Fatal Bus Error

(R/W)

DMAIN_ERIE Write 1 to enable and write 0 to disable EARLY_RECV_INT. (R/W)

DMAIN_FBEE Write 1 to enable and write 0 to disable FATAL_BUS_ERR_INT.(R/W)

DMAIN_ETIE Write 1 to enable and write 0 to disable EARLY_TRANS_INT. (R/W)

DMAIN_RWTE Write 1 to enable and write 0 to disable RECV_WDT_TO_INT. (R/W)

DMAIN_RSE Write 1 to enable and write 0 to disable RECV_PROC_STOP_INT. (R/W)

DMAIN_RBUE Write 1 to enable and write 0 to disable RECV_BUF_UNAVAIL_INT. (R/W)

DMAIN_RIE Write 1 to enable and write 0 to disable RECV_INT. (R/W)

Continued on the next page...

Register 52.8. DMAIN_EN_REG (0x101C)

Continued from the previous page...

DMAIN_UIE Write 1 to enable and write 0 to disable TRANS_UNDFLOW_INT. (R/W)**DMAIN_OIE** Write 1 to enable and write 0 to disable RECV_OVFLOW_INT. (R/W)**DMAIN_TJTE** Write 1 to enable and write 0 to disable TRANS_JABBER_TO_INT. (R/W)**DMAIN_TBUE** Write 1 to enable and write 0 to disable TRANS_BUF_UNAVAIL_INT. (R/W)**DMAIN_TSE** Write 1 to enable and write 0 to disable TRANS_PROC_STOP_INT. (R/W)**DMAIN_TIE** Write 1 to enable and write 0 to disable TRANS_INT. (R/W)**Register 52.9. DMAMISSEDFR_REG (0x1020)**

(reserved)		Overflow_BFOC		Overflow_FC		Overflow_BMFC		Missed_FC	
31	29	28	27		17	16	15		0
0	0	0x0		0x0		0x0		0x0	Reset

Overflow_BFOC Represents the status of the overflow frame counter.

0: No overflow.

1: The overflow frame counter (Overflow_FC) overflows, that is, the RX FIFO overflows with the overflow frame counter at maximum value. In such a scenario, the overflow frame counter is reset to all-zeros.

(R/SS/RC)

Overflow_FC Represents the number of frames missed by the application.

This counter is incremented each time the MTL FIFO overflows. (R/SS/RC)

Overflow_BMFC Represents the status of the missed frame counter.

0: No overflow.

1: The missed frame counter (Missed_FC) overflows, that is, the Host Receive Buffer being unavailable with the missed frame counter at maximum value. In such a scenario, the Missed frame counter is reset to all-zeros.

(R/SS/RC)

Missed_FC Represents the number of frames missed by the controller because of the Host Receive Buffer being unavailable.

This counter is incremented each time the DMA discards an incoming frame. (R/SS/RC)

Register 52.10. DMARINTWDTIMER_REG (0x1024)

(reserved)																RWTC																	
31																8	7	0															
0 0																0x000																Reset	

RWTC Configures the number of system clock cycles multiplied by 256.

The watchdog timer gets triggered with the programmed value after the RX DMA completes the transfer of a frame for which the RI (RECV_INT) status bit is not set because of the setting in the corresponding descriptor RDES1[31].

When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of the automatic setting of RI as per RDES1[31] of any received frame. (R/W)

Register 52.11. DMAAHBSTATUS_REG (0x102C)

(reserved)																																AHB_ST																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
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AHB_ST Configures the state of the AHB master interface.

0: Idle

1: Active

(R/W)

Register 52.12. DMATXCURRDESC_REG (0x1048)

31																															0
0x00000000																															
Reset																															

TRANS_DECR_ADDR_PTR Represents the address of the current transmit descriptor list updated by the DMA during operation.

This field is cleared on reset. (RO)

Register 52.13. DMARXCURREDESC_REG (0x104C)

31	0
0x00000000	
Reset	

RECV_DECR_ADDR_PTR Represents the address of the current receive descriptor list updated by the DMA during operation.

This field is cleared on reset. (RO)

Register 52.14. DMATXCURRADDR_BUF_REG (0x1050)

31	0
0x00000000	
Reset	

TRANS_BUFF_ADDR_PTR Represents the address of the Transmit Buffer updated by the DMA during operation.

This field is cleared on reset. (RO)

Register 52.15. DMARXCURRADDR_BUF_REG (0x1054)

31	0
0x00000000	
Reset	

RECV_BUFF_ADDR_PTR Represents the address of the Receive Buffer updated by the DMA during operation.

This field is cleared on reset. (RO)

Register 52.16. EMACCONFIG_REG (0x0000)

(reserved)		SAIRC	ASS2KP		(reserved)		EMACCRC_STRIP		(reserved)		EMACWATCHDOG		EMACJABBER		(reserved)		EMACJUMBOFRAME		EMACINTERFRAMEGAP		EMACDISABLECRS		EMACMI		EMACFESPEED		EMACRXDOWN		EMACLOOPBACK		EMACRXIPCOFFLOAD		(reserved)		EMACPADCRCSTRIP		EMACBACKOFFLIMIT		EMACTX		EMACRX		PLTF	
31	30	28	27	26	25	24	23	22	21	20	19	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0															
0	0x0		0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0x0	0	0	0	0	0x0	Reset														

SAIRC Configures the source address insertion or replacement for all transmitted frames.

Bit 30 specifies which MAC Address register (0 or 1) is used for source address insertion or replacement based on the values of Bits [29:28].

Bits [29:28]:

0, 1: The input mti_sa_ctrl_i and ati_sa_ctrl_i control the SA field generation

2: Insert the content of the MAC Address register (0 or 1) in the SA field of all transmitted frames

3: Replace the the content of MAC Address register (0 or 1) in the SA field of all transmitted frames

Bit 30:

0: Use MAC Address 0 register

1: Use MAC Address 1 register

(R/W)

ASS2KP Configures whether the MAC considers received frames of more than 2000 bytes as normal packets or giant frames.

0: Giant frames. In this case, if Bit[20] (JE) is 0, the MAC considers all received frames of size more than 1,518 bytes (1,522 bytes for tagged) as Giant frames.

1: Normal packets.

(R/W)

EMACCRC_STRIP Configures whether to strip and drop the last 4 bytes (FCS) of all frames of Ether type (Length/Type field greater than or equal to 1,536) before forwarding the frame to the application.

0: Not strip and drop

1: Strip and drop

(R/W)

EMACWATCHDOG Configures whether to disable the watchdog timer.

0: Enable. In this case, the MAC does not allow a receive frame with more than 2,048 bytes (10,240 if JE is set high) or the value programmed in Watchdog Timeout Register. The MAC cuts off any bytes received after the watchdog limit number of bytes.

1: Disable. In this case, the MAC can receive frames of up to 16,384 bytes.

(R/W)

EMACJABBER Configures whether to disable the jabber timer on the transmitter.

0: Enable. In this case, the MAC cuts off the transmitter if the application sends out more than 2,048 bytes of data (10,240 if JE is set high) during transmission.

1: Disable. In this case, the MAC can transfer frames of up to 16,384 bytes.

(R/W)

Continued on the next page...

Register 52.16. EMACCONFIG_REG (0x0000)

Continued from the previous page...

EMACJUMBOFRAME Configures whether the MAC allows Jumbo frames of 9,018 bytes (9,022 bytes for VLAN tagged frames) without reporting a giant frame error in the receive frame status.

0: Not allow

1: Allow

(R/W)

EMACINTERFRAMEGAP Configures the minimum interframe gap (IFG) between frames during transmission.

0: 96 bit times

1: 88 bit times

2: 80 bit times

3: 72 bit times

4: 64 bit times

5: 56 bit times

6: 48 bit times

7: 40 bit times

In the half-duplex mode, the minimum IFG can be configured only for 64 bit times (IFG = 100).

(R/W)

EMACDISABLECRS Configures whether to disable Carrier Sense during transmission.

0: The MAC transmitter generates Loss of Carrier or No Carrier errors because of Carrier Sense and can even abort the transmissions.

1: The MAC transmitter ignores the MII CRS signal during frame transmission in the half-duplex mode. This request results in no errors generated because of Loss of Carrier or No Carrier during such transmission.

(R/W)

EMACMII Configures the Ethernet line speed.

0: For 1000 Mbps operations

1: For 10 or 100 Mbps operations

In 10 or 100 Mbps operations, the exact line speed is determined by both this bit and the FES (EMACFESPEED) bit. (R/W)

EMACFESPEED Configures the speed in the MII and RMII interface.

0: 10 Mbps

1: 100 Mbps

(R/W)

EMACRXOWN Configures whether to disable receive own.

0: The MAC receives all packets that are given by the PHY while transmitting.

1: the MAC disables the reception of frames when TX_EN triggers an interrupt in the half-duplex mode.

This bit is not applicable if the MAC is operating in the full-duplex mode. (R/W)

Continued on the next page...

Register 52.16. EMACCONFIG_REG (0x0000)

Continued from the previous page...

EMACLOOPBACK Configures whether to enable the loopback mode.

0: Disable.

1: Enable. In this case, the MII Receive clock input (CLK_RX) is required for the loopback to work properly, because the Transmit clock is not looped-back internally.

(R/W)

EMACDUPLEX Configures whether to enable full-duplex mode, so that the MAC can transmit and receive simultaneously.

0: Disable

1: Enable

This bit is RO with a default value of 1 in the full-duplex-only configuration. (R/W)

EMACRXIPCOFFLOAD Configures whether to enable the checksum offload function

0: Disable

1: Enable

When the checksum offload function is enabled, the MAC calculates the 16-bit one's complement of the one's complement sum of all received Ethernet frame payloads. It also checks whether the IPv4 Header checksum (assumed to be bytes 25/26 or 29/30 (VLAN-tagged) of the received Ethernet frame) is correct for the received frame and gives the status in the receive status word. The MAC also appends the 16-bit checksum calculated for the IP header datagram payload (bytes after the IPv4 header) and appends it to the Ethernet frame transferred to the application (when Type 2 COE is deselected). (R/W)

EMACRETRY Configures whether to disable retry, so that the MAC attempts only one transmission, and when a collision occurs on the MII interface the MAC ignores the current frame transmission and reports a Frame Abort with excessive collision error in the transmit frame status.

0: Enable

1: Disable

Valid only in the half-duplex mode. (R/W)

EMACPADCRCSTRIP Configures whether to enable automatic Pad or CRC stripping.

0: Disable. In this case, the MAC passes all incoming frames, without modifying them, to the Host.

1: Enable. In this case, the MAC strips the Pad or FCS field on the incoming frames only if the value of the length field is less than 1,536 bytes. All received frames with length field greater than or equal to 1,536 bytes are passed to the application without stripping the Pad or FCS field.

(R/W)

Continued on the next page...

Register 52.16. EMACCONFIG_REG (0x0000)

Continued from the previous page...

EMACBACKOFFLIMIT Configures the back-off limit determining the random integer number (r) of slot time delays (4,096 bit times for 1000 Mbps and 512 bit times for 10/100 Mbps) for which the MAC waits before rescheduling a transmission attempt during retries after a collision.

00: $k = \min(n, 10)$

01: $k = \min(n, 8)$

10: $k = \min(n, 4)$

11: $k = \min(n, 1)$

where n = retransmission attempt.

The random integer r takes the value in the range $0 \leq r < k$ th power of 2

Valid only in the half-duplex mode. (R/W)

EMACDEFERRALCHECK Configures whether to enable the deferral check function.

0: Disable

1: Enable

(R/W)

EMACTX Configures whether to enable the transmit state machine of the MAC.

0: Disabled after the completion of the transmission of the current frame, and does not transmit any further frames

1: Enabled for transmission on the MII

(R/W)

EMACRX Configures whether to enable the receiver state machine of the MAC. 0: Disabled after the completion of the reception of the current frame, and does not receive any further frames from the MII

1: Enabled for receiving frames from the MII

(R/W)

PLTF Configures the number of preamble bytes that are added to the beginning of every Transmit frame.

0: 7 bytes of preamble

1: 5 byte of preamble

2: 3 bytes of preamble

3: Reserved

The preamble reduction occurs only when the MAC is operating in the full-duplex mode. (R/W)

Register 52.17. EMACFF_REG (0x0004)

RECEIVE_ALL		(reserved)											VTFE			(reserved)				SAFE	SAIF	PCF	DBF	PAM	DAIF	(reserved)		PMODE		
31	30												17	16	15					10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

Reset

RECEIVE_ALL Configures whether to pass all received modules.

0: The MAC Receiver module passes only those frames to the Application that pass the SA or DA address filter.

1: The MAC Receiver module passes all received frames, irrespective of whether they pass the address filter or not, to the Application. The result of the SA or DA filtering is updated (pass or fail) in the corresponding bits in the Receive Status Word.

(R/W)

VTFE Configures whether to enable the VLAN Tag filter.

0: Disable. In this case, the MAC forwards all frames irrespective of the match status of the VLAN Tag.

1: Enable. In this case, the MAC drops VLAN-tagged frames that do not match the VLAN Tag comparison.

(R/W)

SAFE Configures whether to enable the source address filter.

0: Disable. In this case, the MAC forwards the received frame to the application with an updated SAF bit of the RX Status depending on the SA address comparison.

1: Enable. In this case, the MAC compares the SA field of the received frames with the values programmed in the enabled SA registers. If the comparison fails, the MAC drops the frame.

(R/W)

SAIF Configures whether to enable SA inverse filtering.

0: Disable. In this case, frames whose SA does not match the SA registers are marked as failing the SA Address filter.

1: Enable. In this case, the Address Check block operates in inverse filtering mode for the SA address comparison. The frames whose SA matches the SA registers are marked as failing the SA Address filter.

(R/W)

Continued on the next page...

Register 52.17. EMACFF_REG (0x0004)

Continued from the previous page...

PCF Configures the forwarding of all control frames (including unicast and multicast PAUSE frames).

0: The MAC filters all control frames.

1: The MAC forwards all control frames except PAUSE control frames to application even if they fail the Address filter.

2: The MAC forwards all control frames to application even if they fail the Address Filter.

3: The MAC forwards control frames that pass the Address Filter.

The following conditions should be true for the PAUSE control frames processing:

Condition 1: The MAC is in the full-duplex mode and flow control is enabled by setting Bit 2 (RFE) of Register 6 (Flow Control Register) to 1.

Condition 2: The destination address (DA) of the received frame matches the special multicast address or EMACADDRO when Bit 3 (UP) of the Register 6 (Flow Control Register) is set.

Condition 3: The Type field of the received frame is 0x8808 and the OP CODE field is 0x0001.

(R/W)

DBF Configures whether to disable broadcast frames.

0: Enable. The AFM (Address Filtering Module) module passes all received broadcast frames.

1: Disable. The AFM module filters all incoming broadcast frames. In addition, it overrides all other filter settings.

(R/W)

PAM Configures whether to pass all multicast frames.

0: Filtering of multicast frame depends on HMC bit

1: All received frames with a multicast destination address (first bit in the destination address field is '1') are passed

(R/W)

DAIF Configures whether to enable DA inverse filtering for both unicast and multicast frames.

0: Disable

1: Enable

(R/W)

PMODE Configures whether to enable the promiscuous mode.

0: Disable.

1: Enabled. In this case, the Address Filter module passes all incoming frames regardless of its destination or source address. The SA or DA Filter Fails status bits of the Receive Status Word are always cleared when PR (PRI_RATIO) is set.

(R/W)

Register 52.18. EMACMIIADDR_REG (0x0010)

(reserved)																MIIDEV				MIIREG				MIICSRCLK				MIWRITE MIIBUSY																			
31																16				15				11				10				6				5				2				1		0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0x00				0x00				0x00				0		0		Reset															

MIIDEV Configures which of the 32 possible PHY devices are being accessed. (R/W)

MIIREG Configures the desired address register in the selected PHY device. (R/W)

MIICSRCLK Configures the CSR clock frequency.

0: APB clock frequency is 80 MHz, and MDC clock frequency is APB_CLK/42

3: APB clock frequency is 40 MHz, and MDC clock frequency is APB_CLK/26

Other values are reserved

(R/W)

MIIWRITE Configures the direction of the operation that uses [MII_DATA](#).

0: Read operation

1: Write operation

(R/W)

MIIBUSY Configures the state of the MII.

0: Idle

1: Busy

This bit is used in combination with [MIIREG](#) and [MII_DATA](#).

This bit should read logic 0 (default) before writing to [MIIREG](#) and [MII_DATA](#).

To read or write to [MIIREG](#) and [MII_DATA](#), software (the user) should set this field to 1.

[MII_DATA](#) should be kept valid (data remains unchanged) when it is accessed until this field is cleared by hardware (the MAC).

Note that ESP32-P4 MAC does not receive ACK from PHY during a read or write access to [MIIREG](#) and [MII_DATA](#). (R/WS/SC)

Register 52.19. EMACMIIDATA_REG (0x0014)

(reserved)																MII_DATA																	
31																16	15																0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x00000															Reset		

MII_DATA Configures the 16-bit data value read from the PHY after a Management Read operation or the 16-bit data value to be written to the PHY before a Management Write operation. (R/W)

Register 52.20. EMACFC_REG (0x0018)

PAUSE_TIME																(reserved)								DZPQ		(reserved)		PLT		UPFD		RFCE		TFCE		FCBBA																																																																																																																																																																																																											
31																16																15																8																7																6																5																4																3																2																1																0																																																															
0x00000																0																0																0																0																0																0																0																0																0x0																0																0																0																0																Reset															

PAUSE_TIME Configures the value to be used in the Pause Time field in the transmit control frame. If the Pause Time bits is configured to be double-synchronized to the (G)MII clock domain, then consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain. (R/W)

DZPQ Configures whether to disable the automatic generation of the Zero-Quanta Pause Control frames.
 0: Enable
 1: Disable
 (R/W)

PLT Configures the threshold of the PAUSE timer at which the PAUSE Frame is automatically retransmitted.
 0: The threshold is Pause time minus 4 slot times (PT - 4 slot times).
 1: The threshold is Pause time minus 28 slot times (PT - 28 slot times).
 2: The threshold is Pause time minus 144 slot times (PT - 144 slot times).
 3: The threshold is Pause time minus 256 slot times (PT - 256 slot times).
 The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot-times), and PLT = 01, then a second PAUSE frame is automatically transmitted 228 (256 - 28) slot times after the first PAUSE frame is transmitted.
 (R/W)

UPFD Configures whether to enable unicast pause frame detection.
 0: Disable. The MAC only detects Pause frames with a unique multicast address.
 1: Enable. The MAC can also detect Pause frames with unicast address of the station. This unicast address should be as specified in the EMACADDRO High Register and EMACADDRO Low Register.
 (R/W)

RFCE Configures whether to enable the decode function of the Pause time.
 0: Disable
 1: Enable. The MAC decodes the received Pause frame and disables its transmitter for a specified (Pause) time
 (R/W)

TFCE Configures whether to enable transmit flow control in full-duplex mode, and whether to enable the back-pressure feature in half-duplex mode.
 0: Disable
 1: Enable
 (R/W)

Continued on the next page...

Register 52.20. EMACFC_REG (0x0018)

Continued from the previous page...

FCBBA Configures whether to enable a Pause Control frame in the full-duplex mode, and whether to activate the back-pressure function in the half-duplex mode if the TFCE bit is set.

0: Disable

1: Enable

In the full-duplex mode, this bit should be read as 0 before writing to the Flow Control register. To initiate a Pause control frame, the Application must set this bit to 1. During a transfer of the Control Frame, this bit continues to be set to signify that a frame transmission is in progress. After the completion of Pause control frame transmission, the MAC resets this bit to 0. The Flow Control register should not be written to until this bit is cleared. (R/WS/SC)

In the half-duplex mode, when this bit is set (and TFCE is set), then backpressure is asserted by the MAC. During backpressure, when the MAC receives a new frame, the transmitter starts sending a JAM pattern resulting in a collision. When the MAC is configured for the full-duplex mode, the BPA is automatically disabled. (R/W)

Register 52.21. EMACVLANTAG_REG (0x001C)

(reserved)																ESVL VTIM ETV			VL																
31																19	18	17	16	15															0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0x0000															Reset	

ESVL Configures whether to enable S-VLAN.

0: Disable.

1: Enable. The MAC transmitter and receiver also consider the S-VLAN (Type = 0x88A8) frames as valid VLAN tagged frames.

(R/W)

VTIM Configures whether to enable VLAN Tag inverse match.

0: Disable. The frames with matched VLAN Tag are marked as matched.

1: Enable. The frames that do not have matching VLAN Tag are marked as matched.

(R/W)

ETV Configures whether to enable 12-bit VLAN Tag comparison.

0: Disable. All 16 bits of the 15th and 16th bytes of the received VLAN frame are used for comparison and VLAN hash filtering.

1: Enable. A 12-bit VLAN identifier is used for comparing and filtering instead of the complete 16-bit VLAN tag. Bits [11:0] of VLAN tag are compared with the corresponding field in the received VLAN-tagged frame.

(R/W)

VL Configures the 802.1Q VLAN tag to identify the VLAN frames and is compared to the 15th and 16th bytes of the frames being received for VLAN frames.

When the ETV bit is set, only the VID (Bits[11:0]) is used for comparison.

If VL (VL[11:0] if ETV is set) is all zeros, the MAC does not check the fifteenth and 16th bytes for VLAN tag comparison, and declares all frames with a Type field value of 0x8100 or 0x88a8 as VLAN frames.(R/W)

Register 52.22. EMACDEBUG_REG (0x0024)

(reserved)							MTLTSFFS	MTLTFNES	(reserved)		MTLTFWCS	MTLTFRCS	MACTP	MACTFCS	MACTPES	(reserved)							MTLRFFLS	(reserved)		MTLRFRCs	MTLRFWCAS	(reserved)		MACRFFCS	MACRPES		
31						26	25	24	23	22	21	20	19	18	17	16	15						10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0x0	0	0x0	0	0	0	0	0	0	0	0	0x0	0	0	0x0	0	0	0	0x0	0	0	Reset	

MTLTSFFS Represents whether MTL TX Status FIFO is full and can accept any more frames for transmission.

0: Not Full

1: Full

(RO)

MTLTFNES Represents whether MTL TX FIFO is not empty and some data is left for transmission.

0: Empty

1: Not empty

(RO)

MTLTFWCS Represents whether MTL TX FIFO Write Controller is active and transferring data to the TX FIFO.

0: Inactive

1: Active

(RO)

MTLTFRCS Represents the state of the TX FIFO Read Controller.

0: IDLE state

1: READ state (transferring data to MAC transmitter)

2: Waiting for TXStatus from MAC transmitter

3: Writing the received TXStatus or flushing the TX FIFO

(RO)

MACTP Represents whether the MAC transmitter is in the PAUSE condition (in the full-duplex only mode) and hence does not schedule any frame for transmission.

0: Not in pause

1: In pause

(RO)

MACTFCS Represents the state of the MAC Transmit Frame Controller module.

0: IDLE state

1: Waiting for Status of the previous frame or IFG or backoff period to be over

2: Generating and transmitting a PAUSE control frame (in the full-duplex mode)

3: Transferring input frame for transmission

(RO)

MACTPES Represents the status of the MAC MII transmit protocol engine.

0: IDLE state

1: Actively transmitting data

(RO)

Continued on the next page...

Register 52.22. EMACDEBUG_REG (0x0024)

Continued from the previous page...

MTLRFFLS Represents the status of the fill-level of the RX FIFO.

- 0: RX FIFO Empty
 - 1: RX FIFO fill level is below the flow-control deactivate threshold
 - 2: RX FIFO fill level is above the flow-control activate threshold
 - 3: RX FIFO Full
- (RO)

MTLRFRCs Represents the state of the RX FIFO read Controller.

- 0: IDLE state
 - 1: Reading frame data
 - 2: Reserved
 - 3: Flushing the frame data and status
- (RO)

MTLRFWCAS Represents the state of the MTL RX FIFO Write Controller.

- 0: Inactive
 - 1: Active and is transferring a received frame to the FIFO.
- (RO)

MACRFFCS Represents the state of the small FIFO Read (bit 1) and Write (bit 0) controllers of the MAC Receive Frame Controller Module.

- 0: Inactive
 - 1: Active
- (RO)

MACRPES Represents the state of the MAC GMII or MII receive protocol engine.

- 0: IDLE
 - 1: Actively receiving data
- (RO)

Register 52.23. PMT_RWUFFR_REG (0x0028)



WKUPPKTFILTER When RWKPTR is 0 ~ 3, configures the masking of filter 0 ~ 3 wakeup frame.

Bit[30:0] corresponds to offset+j (j = 0 ~ 30).

Bit[31] has to be 0. 0: Unmask

1: Mask

When RWKPTR is 4, bit[3]/bit[11]/bit[19]/bit[27] configure the target address type, and bit[0]/bit[8]/bit[16]/bit[24] configure whether to enable filter 0 ~ 3.

0: Unicast/Disable

1: Multicast/Enable

When RWKPTR is 5, configures the offset of the filter 0 ~ 3 masked byte.

Bit[7:0] configures the offset of filter 0 masked byte, and [31:24] configures the offset of filter 3 masked byte.

When RWKPTR is 6 ~ 7, configures the 16-bit CRC value of filter 0 ~ 3 masked byte starting from offset based on the formula:

$$G(x) = x^{16} + x^{15} + x^2 + 1$$

Bit[7:0] configures the CRC value of filter 0 or filter 2 masked byte when RWKPTR is 6 or 7 respectively, and bit[31:16] configures the CRC value of filter 1 or filter 3 masked byte when RWKPTR is 6 or 7 respectively.

(R/W)

Register 52.24. PMT_CSR_REG (0x002C)

RWKFILTRST (reserved)				RWKPTR										(reserved)										GLBLUCAST (reserved)				RWKPRCVD MGKPRCVD		(reserved)		RWKPKTEN MGKPKTEN		PWRDWN	
31	30	29	28	24										23	10										9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset		

Reset

RWKFILTRST Configures whether to reset the RWKPTR register.

0: Release from reset

1: Reset

(R/WS/SC)

RWKPTR Configures the purpose of the PMT_RWUFR register. For details, see the description of PMT_RWUFR. (RO)

GLBLUCAST Configures whether to enable the unicast frame of target address filter as the wakeup frame.

0: Disable

1: Enable

(R/W)

RWKPRCVD Represents whether a remote wakeup frame event has been received.

0: Not received

1: Received

(R/SS/RC)

MGKPRCVD Represents whether a magic frame event has been received.

0: Not received

1: Received

(R/SS/RC)

RWKPKTEN Configures whether to enable power management events using remote wakeup frames.

0: Disable

1: Enable

(R/W)

MGKPKTEN Configures whether to enable power management events using magic frames.

0: Disable

1: Enable

(R/W)

PWRDWN Configures whether the EMAC receiver drops all received frames until it receives the expected magic frame or remote wakeup frame.

Valid only when MGKPKTEN, GLBLUCAST or RWKPKTEN is 1. (R/WS/SC)

Register 52.25. EMACLPI_CSR_REG (0x0030)

(reserved)												LPITXA (reserved) PLS LPIEN				(reserved)				RLPIST TLPIST		(reserved)		RLPIEX RLPIEN TLPIEX TLPIEN						
31												20	19	18	17	16	15	10				9	8	7	4		3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

Reset

LPITXA Configures the behavior of the MAC when it is entering or coming out of the LPI mode on the transmit side.

0: This bit directly controls behavior of the MAC when it is entering or coming out of the LPI mode.

1: This bit controls behavior of the MAC when it is entering or coming out of the LPI mode in combination with LPIEN. When LPIEN is also 1, the MAC enters the LPI mode only after all outstanding frames (in the core) and pending frames (in the application interface) have been transmitted. The MAC comes out of the LPI mode when the application sends any frame for transmission

(R/W)

PLS Configures the link status of the PHY.

0: Link up

1: Link down

(R/W)

LPIEN Configures whether the EMAC transmitter enters the LPI state.

0: Exit

1: Enter

This bit is cleared when the LPITXA bit is set and the MAC exits the LPI state because of the arrival of a new packet for transmission.

(R/W/SC)

RLPIST Represents the receive LPI state.

0: Inactive

1: The EMAC is receiving the LPI pattern

(RO)

TLPIST Represents the transmit LPI state.

0: Inactive

1: The EMAC is transmitting the LPI pattern

(RO)

RLPIEX Represents the receive LPI exit state.

0: Inactive

1: The EMAC Receiver has stopped receiving the LPI pattern, exited the LPI state, and resumed the normal reception

(R/SS/RC)

Continued on the next page...

Register 52.25. EMACLPI_CSR_REG (0x0030)

Continued from the previous page...

- RLPIEN** Represents the receive LPI entry state.
0: Inactive
1: The EMAC Receiver has received an LPI pattern and entered the LPI state (R/SS/RC)
- TLPIEX** Represents the transmit LPI exit state.
0: Inactive
1: The EMAC transmitter has exited the LPI state after the user has cleared the LPIEN bit and the LPI_TW_TIMER has expired (R/SS/RC)
- TLPIEN** Represents the transmit LPI entry state.
0: Inactive
1: The EMAC Transmitter has entered the LPI state because of the setting of the LPIEN bit (R/SS/RC)

Register 52.26. EMACLPITIMERSCONTROL_REG (0x0034)

(reserved)						LPI_LS_TIMER						LPI_TW_TIMER																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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- LPI_LS_TIMER** Configures the minimum time for which the link status from the PHY should be up.
Measurement unit: millisecond.
The default value is 1000 (1 second) as defined in the IEEE standard.
(R/W)
- LPI_TW_TIMER** Configures the minimum time for which the MAC waits after it stops transmitting the LPI pattern to the PHY and before it resumes the normal transmission.
Measurement unit: millisecond.
The TLPIEX status bit is set after the expiry of this timer.
(R/W)

Register 52.27. EMACINTS_REG (0x0038)

(reserved)											LPIINTS TINTS		(reserved)		PMTINTS (reserved)			
31											11	10	9	8	5	3	2	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

Reset

LPIINTS The raw interrupt status of EMAC_LPI_INT. (RO)**TINTS** The raw interrupt status of TS_TRI_INT. (R/SS/RC)**PMTINTS** The raw interrupt status of EMAC_PMP_INT. (RO)**Register 52.28. EMACINTMASK_REG (0x003C)**

(reserved)											LPIINTMASK TSINTMASK		(reserved)		PMTINTMASK (reserved)						
31											11		10	9	8	4		3	2	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

Reset

LPIINTMASK Write 1 to mask EMAC_LPI_INT.(R/W)**TSINTMASK** Write 1 to mask TS_TRI_INT. (R/W)**PMTINTMASK** Write 1 to mask EMAC_PMP_INT. (R/W)**Register 52.29. EMACADDRHIGH_REG (0x0040)**

ADDRESS_ENABLE0																(reserved)																MAC_ADDRESS0_HI															
31	30															16	15															0															
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x0FFFF															Reset																

Reset

ADDRESS_ENABLE0 This bit is always set to 1. (RO)**MAC_ADDRESS0_HI** Configures the upper 16 bits of the first 6-byte MAC address [47:32].

The MAC uses this field for filtering the received frames and inserting the MAC address in the Transmit Flow Control (PAUSE) Frames. (R/W)

Register 52.30. EMACADDR0LOW_REG (0x0044)

31	0
0xFFFFFFFF	
Reset	

EMACADDR0LOW_REG Configures the lower 32 bits of the first 6-byte MAC address.

The MAC uses this field for filtering the received frames and inserting the MAC address in the Transmit Flow Control (PAUSE) Frames. (R/W)

Register 52.31. EMACADDR1HIGH_REG (0x0048)

ADDRESS_ENABLE1 SOURCE_ADDRESS		MASK_BYTE_CONTROL		(reserved)								MAC_ADDRESS1_HI							
31	30	29	24	23						16	15								0
0	0		0x00	0	0	0	0	0	0	0	0							0x0000	Reset

ADDRESS_ENABLE1 Configures whether to enable perfect filtering using the second MAC address.

0: Disable

1: Enable

(R/W)

SOURCE_ADDRESS Configures whether to which fields of the received frame EMACADDR1 [47:0] compares.

0: The DA fields

1: The SA fields

(R/W)

MASK_BYTE_CONTROL Configures whether to mask MAC address byte bytes when comparing the received DA or SA with the contents of EMACADDR1.

0: Unmask

1: Mask

The correspondence between the mask control bits and the bytes:

Bit[29]: EMACADDR1HIGH_REG[15:8]

Bit[28]: EMACADDR1HIGH_REG[7:0]

Bit[27]: EMACADDR1LOW_REG[31:24]

Bit[24]: EMACADDR1LOW_REG[7:0]

You can filter a group of addresses (known as group address filtering) by masking one or more bytes of the address.(R/W)

MAC_ADDRESS1_HI Configures the upper 16 bits of the second 6-byte MAC address [47:32]. (R/W)

Register 52.32. EMACADDR1LOW_REG (0x004C)

31	0
0xFFFFFFFF	
Reset	

EMACADDR1LOW_REG Configures the lower 32 bits of the second 6-byte MAC address.
 The content of this field is undefined until loaded by the Application after the initialization process. (R/W)

Register 52.33. EMACADDR2HIGH_REG (0x0050)

ADDRESS_ENABLE2 SOURCE_ADDRESS2		MASK_BYTE_CONTROL2		(reserved)		MAC_ADDRESS2_HI																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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ADDRESS_ENABLE2 Configures whether to enable perfect filtering using the third MAC address.
 0: Disable
 1: Enable
 (R/W)

SOURCE_ADDRESS2 Configures whether to which fields of the received frame EMACADDR2 [47:0] compares.
 0: The DA fields
 1: The SA fields
 (R/W)

MASK_BYTE_CONTROL2 Configures whether to mask MAC address byte bytes when comparing the received DA or SA with the contents of EMACADDR2.

0: Unmask
 1: Mask

The correspondence between the mask control bits and the bytes:

Bit[29]: EMACADDR2HIGH_REG[15:8]

Bit[28]: EMACADDR2HIGH_REG[7:0]

Bit[27]: EMACADDR2LOW_REG[31:24]

Bit[24]: EMACADDR2LOW_REG[7:0]

You can filter a group of addresses (known as group address filtering) by masking one or more bytes of the address.(R/W)

MAC_ADDRESS2_HI Configures the upper 16 bits of the third 6-byte MAC address [47:32]. (R/W)

Register 52.34. EMACADDR2LOW_REG (0x0054)

31	0
0xFFFFFFFF	
Reset	

EMACADDR2LOW_REG Configures the lower 32 bits of the third 6-byte MAC address.

The content of this field is undefined until loaded by the Application after the initialization process. (R/W)

Register 52.35. EMACADDR3HIGH_REG (0x0058)

ADDRESS_ENABLE3 SOURCE_ADDRESS3		MASK_BYTE_CONTROL3		(reserved)		MAC_ADDRESS3_HI																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
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ADDRESS_ENABLE3 Configures whether to enable perfect filtering using the fourth MAC address.

0: Disable

1: Enable

(R/W)

SOURCE_ADDRESS3 Configures whether to which fields of the received frame EMACADDR3 [47:0] compares.

0: The DA fields

1: The SA fields

(R/W)

MASK_BYTE_CONTROL3 Configures whether to mask MAC address byte bytes when comparing the received DA or SA with the contents of EMACADDR3.

0: Unmask

1: Mask

The correspondence between the mask control bits and the bytes:

Bit[29]: EMACADDR3HIGH_REG[15:8]

Bit[28]: EMACADDR3HIGH_REG[7:0]

Bit[27]: EMACADDR3LOW_REG[31:24]

Bit[24]: EMACADDR3LOW_REG[7:0]

You can filter a group of addresses (known as group address filtering) by masking one or more bytes of the address. (R/W)

MAC_ADDRESS3_HI Configures the upper 16 bits of the fourth 6-byte MAC address [47:32]. (R/W)

Register 52.36. EMACADDR3LOW_REG (0x005C)

31	0
0xFFFFFFFF	
Reset	

EMACADDR3LOW_REG Configures the lower 32 bits of the fourth 6-byte MAC address.

The content of this field is undefined until loaded by the Application after the initialization process. (R/W)

Register 52.37. EMACADDR4HIGH_REG (0x0060)

ADDRESS_ENABLE4 SOURCE_ADDRESS4																MASK_BYTE_CONTROL4								(reserved)																MAC_ADDRESS4_HI															
31	30	29	24																23	16																15	0																		
0	0	0x00																0	0	0	0	0	0	0	0	0	0x0FFF																Reset												

ADDRESS_ENABLE4 Configures whether to enable perfect filtering using the fifth MAC address.

0: Disable

1: Enable

(R/W)

SOURCE_ADDRESS4 Configures whether to which fields of the received frame EMACADDR4 [47:0] compares.

0: The DA fields

1: The SA fields

(R/W)

MASK_BYTE_CONTROL4 Configures whether to mask MAC address byte bytes when comparing the received DA or SA with the contents of EMACADDR4.

0: Unmask

1: Mask

The correspondence between the mask control bits and the bytes:

Bit[29]: EMACADDR4HIGH_REG[15:8]

Bit[28]: EMACADDR4HIGH_REG[7:0]

Bit[27]: EMACADDR4LOW_REG[31:24]

Bit[24]: EMACADDR4LOW_REG[7:0]

You can filter a group of addresses (known as group address filtering) by masking one or more bytes of the address.(R/W)

MAC_ADDRESS4_HI Configures the upper 16 bits of the fifth 6-byte MAC address [47:32]. (R/W)

Register 52.38. EMACADDR4LOW_REG (0x0064)

31	0
0xFFFFFFFF	
Reset	

EMACADDR4LOW_REG Configures the lower 32 bits of the fifth 6-byte MAC address.

The content of this field is undefined until loaded by the Application after the initialization process. (R/W)

Register 52.39. EMACADDR5HIGH_REG (0x0068)

ADDRESS_ENABLE5 SOURCE_ADDRESS5		MASK_BYTE_CONTROL5		(reserved)		MAC_ADDRESS5_HI																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
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ADDRESS_ENABLE5 Configures whether to enable perfect filtering using the sixth MAC address.

0: Disable

1: Enable

(R/W)

SOURCE_ADDRESS5 Configures whether to which fields of the received frame EMACADDR5 [47:0] compares.

0: The DA fields

1: The SA fields

(R/W)

MASK_BYTE_CONTROL5 Configures whether to mask MAC address byte bytes when comparing the received DA or SA with the contents of EMACADDR5.

0: Unmask

1: Mask

The correspondence between the mask control bits and the bytes:

Bit[29]: EMACADDR5HIGH_REG[15:8]

Bit[28]: EMACADDR5HIGH_REG[7:0]

Bit[27]: EMACADDR5LOW_REG[31:24]

Bit[24]: EMACADDR5LOW_REG[7:0]

You can filter a group of addresses (known as group address filtering) by masking one or more bytes of the address.(R/W)

MAC_ADDRESS5_HI Configures the upper 16 bits of the sixth 6-byte MAC address [47:32]. (R/W)

Register 52.40. EMACADDR5LOW_REG (0x006C)

31	0
0xFFFFFFFF	
Reset	

EMACADDR5LOW_REG Configures the lower 32 bits of the sixth 6-byte MAC address.

The content of this field is undefined until loaded by the Application after the initialization process. (R/W)

Register 52.41. EMACADDR6HIGH_REG (0x0070)

ADDRESS_ENABLE6 SOURCE_ADDRESS6		MASK_BYTE_CONTROL6		(reserved)		MAC_ADDRESS6_HI																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
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ADDRESS_ENABLE6 Configures whether to enable perfect filtering using the seventh MAC address.

0: Disable

1: Enable

(R/W)

SOURCE_ADDRESS6 Configures whether to which fields of the received frame EMACADDR 6[47:0] compares.

0: The DA fields

1: The SA fields

(R/W)

MASK_BYTE_CONTROL6 Configures whether to mask MAC address byte bytes when comparing the received DA or SA with the contents of EMACADDR6.

0: Unmask

1: Mask

The correspondence between the mask control bits and the bytes:

Bit[29]: EMACADDR6HIGH_REG[15:8]

Bit[28]: EMACADDR6HIGH_REG[7:0]

Bit[27]: EMACADDR6LOW_REG[31:24]

Bit[24]: EMACADDR6LOW_REG[7:0]

You can filter a group of addresses (known as group address filtering) by masking one or more bytes of the address. (R/W)

MAC_ADDRESS6_HI Configures the upper 16 bits of the seventh 6-byte MAC address [47:32]. (R/W)

Register 52.42. EMACADDR6LOW_REG (0x0074)

31	0
0xFFFFFFFF	
Reset	

EMACADDR6LOW_REG Configures the lower 32 bits of the seventh 6-byte MAC address.
 The content of this field is undefined until loaded by the Application after the initialization process. (R/W)

Register 52.43. EMACADDR7HIGH_REG (0x0078)

ADDRESS_ENABLE7 SOURCE_ADDRESS7		MASK_BYTE_CONTROL7		(reserved)								MAC_ADDRESS7_HI		
31	30	29	24	23	16	15	0							
0	0	0x00		0	0	0	0	0	0	0	0	0x0FFF		Reset

ADDRESS_ENABLE7 Configures whether to enable perfect filtering using the eighth MAC address.
 0: Disable
 1: Enable
 (R/W)

SOURCE_ADDRESS7 Configures whether to which fields of the received frame EMACADDR7 [47:0] compares.
 0: The DA fields
 1: The SA fields
 (R/W)

MASK_BYTE_CONTROL7 Configures whether to mask MAC address byte bytes when comparing the received DA or SA with the contents of EMACADDR7.

0: Unmask
 1: Mask

The correspondence between the mask control bits and the bytes:

Bit[29]: EMACADDR7HIGH_REG[15:8]

Bit[28]: EMACADDR7HIGH_REG[7:0]

Bit[27]: EMACADDR7LOW_REG[31:24]

Bit[24]: EMACADDR7LOW_REG[7:0]

You can filter a group of addresses (known as group address filtering) by masking one or more bytes of the address. (R/W)

MAC_ADDRESS7_HI Configures the upper 16 bits of the eighth 6-byte MAC address [47:32]. (R/W)

Register 52.44. EMACADDR7LOW_REG (0x007C)

31	0
0xFFFFFFFF	
Reset	

EMACADDR7LOW_REG Configures the lower 32 bits of the eighth 6-byte MAC address.
 The content of this field is undefined until loaded by the Application after the initialization process. (R/W)

Register 52.45. EMACADDR8HIGH_REG (0x0080)

ADDRESS_ENABLE8 SOURCE_ADDRESS8		MASK_BYTE_CONTROL8		(reserved)		MAC_ADDRESS8_HI																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
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ADDRESS_ENABLE8 Configures whether to enable perfect filtering using the ninth MAC address.
 0: Disable
 1: Enable
 (R/W)

SOURCE_ADDRESS8 Configures whether to which fields of the received frame EMACADDR8 [47:0] compares.
 0: The DA fields
 1: The SA fields
 (R/W)

MASK_BYTE_CONTROL8 Configures whether to mask MAC address byte bytes when comparing the received DA or SA with the contents of EMACADDR8.
 0: Unmask
 1: Mask

The correspondence between the mask control bits and the bytes:

Bit[29]: EMACADDR8HIGH_REG[15:8]

Bit[28]: EMACADDR8HIGH_REG[7:0]

Bit[27]: EMACADDR8LOW_REG[31:24]

Bit[24]: EMACADDR8LOW_REG[7:0]

You can filter a group of addresses (known as group address filtering) by masking one or more bytes of the address. (R/W)

MAC_ADDRESS8_HI Configures the upper 16 bits of the ninth 6-byte MAC address [47:32]. (R/W)

Register 52.46. EMACADDR8LOW_REG (0x0084)

31	0
0x0FFFFFFF	
Reset	

EMACADDR8LOW_REG Configures the lower 32 bits of the ninth 6-byte MAC address.

The content of this field is undefined until loaded by the Application after the initialization process. (R/W)

Register 52.47. EMACSTATUS_REG (0x00D8)

(reserved)																(reserved)				(reserved)				JABBER_TIMEOUT				(reserved)				LINK_MODE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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JABBER_TIMEOUT Represents whether a receive jabber error occurs.

0: Not occur

1: Occur

(RO)

LINK_MODE Represents the current mode of operation of the link.

0: Half-duplex

1: Full-duplex

(RO)

Register 52.48. EMACWDOGTO_REG (0x00DC)

(reserved)																PWDGEN				(reserved)				WDOGTO															
31																17				16				15				14				13				0			
0																0				0				0				0x0000				Reset							

PWDGEN Configures whether the watchdog timeout is programmable.

0: Not programmable. The watchdog timeout for a received frame is controlled by the setting of EMACCONFIG_REG[23] (WD) and EMACCONFIG_REG[20] (JE).

1: Programmable via the WDOGTO field if EMACCONFIG_REG[23] (WD) is 0.

(R/W)

WDOGTO Configures the watchdog timeout for a received frame when PWDGEN is 1 and EMACCONFIG_REG[23] (WD) is 0.

If the length of a received frame exceeds the value of this field, such frame is terminated and declared as an error frame. (R/W)

Register 52.49. EMAXVLTCTRL_REG (0x0584)

(reserved)												CSVL		VLP	VLC	VLT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
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CSVL Configures which type is inserted or replaced in the 13th and 14th bytes of transmitted frames.

0: C-VLAN type (0x8100)

1: S-VLAN type (0x88A8)

(R/W)

VLP Configures what controls VLAN deletion, insertion, or replacement.

0: The control input signal

1: VLC

(R/W)

VLC Configures the VLAN tag in transmit frames.

0: No VLAN tag deletion, insertion, or replacement.

1: VLAN tag deletion. The MAC removes the VLAN type (bytes 13 and 14) and VLAN tag (bytes 15 and 16) of all transmitted frames with VLAN tags.

2: VLAN tag insertion. The MAC inserts VLT in bytes 15 and 16 of the frame after inserting the Type value (0x8100/0x88a8) in bytes 13 and 14. This operation is performed on all transmitted frames, irrespective of whether they already have a VLAN tag.

3: VLAN tag replacement. The MAC replaces VLT in bytes 15 and 16 of all VLAN-type transmitted frames (Bytes 13 and 14 are 0x8100/0x88a8).

(R/W)

VLT Configures the value of the VLAN tag to be inserted or replaced.

The value must only be changed when the transmit lines are inactive or during the initialization phase. (R/W)

Register 52.50. EMACTSTPCTRL_REG (0x0700)

(reserved)												TSENMACADDR		SNAPTYPSEL		TSMSTRENA		TSEVNTENA		TSIPV4ENA		TSIPV6ENA		TSVER2ENA		TSCTRLSSR		TSEWALL		(reserved)		TSADDRREG		TSTRIG		TSUPDT		TSINIT		TSCFUPDT		TSENA	
31												19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0												
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset			

TSENMACADDR Configures whether to enable the DA MAC address (that matches any MAC Address register) for PTP frame filtering when PTP is directly sent over Ethernet.

0: Disable

1: Enable

(R/W)

SNAPTYPSEL Configures the set of PTP packet types for which snapshot needs to be taken, in combination with TSMSTRENA and TSEVNTENA.

SNAPTYPSEL, TSMSTRENA, TSEVNTENA:

00x0: SYNC, Follow_Up, Delay_Req, Delay_Resp

0001: SYNC

0011: Delay_Req

01x0: SYNC, Follow_Up, Delay_Req, Delay_Resp, Pdelay_Req, Pdelay_Resp, Pdelay_Resp_Follow_Up

0101: SYNC, Pdelay_Req, Pdelay_Resp

0111: Delay_Req, Pdelay_Req, Pdelay_Resp

10xx: SYNC, Delay_Req

11xx: Pdelay_Req, Pdelay_Resp

(R/W)

TSMSTRENA Configures whether to enable snapshot only for messages relevant to the master node.

0: Disable. The snapshot is taken for the messages relevant to the slave node

1: Enable

(R/W)

TSEVNTENA Configures whether to enable timestamp snapshot only for event messages (SYNC, Delay_Req, Pdelay_Req, Pdelay_Resp).

0: Disable. The snapshot is taken for all messages except Announce, Management, and Signaling

1: Enable

(R/W)

TSIPV4ENA Configures whether to enable the processing of PTP frames sent over IPv4-UDP.

0: Disable. The MAC ignores the PTP transported over UDP-IPv4 packets

1: Enable. The MAC receiver processes the PTP packets encapsulated in UDP over IPv4 packets

(R/W)

Continued on the next page...

Register 52.50. EMACTSTPCTRL_REG (0x0700)

Continued from the previous page...

TSIPV6ENA Configures whether to enable the processing of PTP frames sent over IPv6-UDP.

0: Disable. The MAC ignores the PTP transported over UDP-IPv6 packets

1: Enable. The MAC receiver processes the PTP packets encapsulated in UDP over IPv6 packets

(R/W)

TSIPENA Configures whether to enable the processing of PTP sent over Ethernet frames.

0: Disable. the MAC ignores the PTP over Ethernet packets

1: Enable. The MAC receiver processes the PTP packets encapsulated directly in the Ethernet frames

(R/W)

TSVER2ENA Configures whether to enable PTP packet processing for version 2 format.

0: Disable. The PTP packets are processed using the version 1 format

1: Enable. The PTP packets are processed using the 1588 version 2 format

(R/W)

TSCTRLSSR Configures the timeout digital or binary rollover.

0: The Timestamp Low register rolls over after the value of the sub-second register 0x7FFF_FFFF. The sub-second increment has to be programmed correctly depending on the PTP reference clock frequency and the value of this bit.

1: The Timestamp Low register rolls over after 0x3B9A_C9FF value (that is, 1 nanosecond accuracy) and increments the timestamp (High) seconds.

(R/W)

TSENALL Configures whether to enable timestamp for all frames received by the MAC.

0: Disable

1: Enable

(R/W)

TSADDREG Configures whether to update the content of the EMACTSTPADDEND_REG register in the PTP block for fine correction.

0: Not update

1: Update

This bit should be zero before setting it. (R/WS/SC)

Continued on the next page...

Register 52.50. EMACTSTPCTRL_REG (0x0700)

Continued from the previous page...

TSTRIG Configures whether to enable the timestamp interrupt trigger.

0: Disable

1: Enable. The timestamp interrupt is generated when the System Time becomes greater than the value written in the Target Time register

(R/WS/SC)

TSUPDT Configures whether to update (add or subtract) the system time with the value specified in EMAC2NDUPT_REG and EMACNSUPT_REG.

0: Not update

1: Update

This bit should be read as zero before updating it.

EMACSYSTIMHIGHWORD2ND_REG is not updated. (R/WS/SC)

TSINIT Configures whether to initialize (overwrite) the system time with the value specified in EMAC2NDUPT_REG and EMACNSUPT_REG.

0: Not initialize

1: Initialize

This bit should be read as zero before updating it.

EMACSYSTIMHIGHWORD2ND_REG will be initialized. (R/WS/SC)

TSCFUPDT Configures the method used to update the system time.

0: Coarse method

1: Find update method

(R/W)

TSENA Configures whether to add the timestamp for the transmit and receive frames.

0: Not add the timestamp, and receive frames and the Timestamp Generator is also suspended

1: Add the timestamp

You need to initialize the Timestamp (system time) after setting this bit to 1.

On the receive side, the MAC processes the 1588 frames only if this bit is set. (R/W)

Register 52.51. EMACSUB2NDINCR_REG (0x0704)

(reserved)																SSINC									
31																8	7	0							
0 0																0x00								Reset	

SSINC Configures the value accumulated every clock cycle (PTP clock) with the contents of the sub-second register.

For example, when PTP clock is 50 MHz (the period is 20 ns), you should program 20 (0x14) when the System Time-Nanoseconds register has an accuracy of 1 ns (EMACTSTPCTRL_REG[9], i.e., the TSCTRLSSR bit is set). When TSCTRLSSR is cleared, the Nanoseconds register has a resolution of 0.465 ns. In this case, you should program a value of 43 (0x2B) that is derived by $20 \text{ ns} / 0.465$. (R/W)

Register 52.52. EMACSYSTM2ND_REG (0x0708)

31																																0	
0x00000000																																	Reset

TSS Represents the current value of the System Time maintained by the MAC.

Measurement unit: second. (RO)

Register 52.53. EMACSYSTMNS_REG (0x070C)

(reserved)																																TSSS																																																																
31	30																																																																																															0
0	0x00000000																																																																Reset																															

TSSS Represents the sub-second representation of time, with an accuracy of 0.46 ns.

When EMACTSTPCTRL_REG[9] (TSCTRLSSR) is set, each bit represents 1 ns and the maximum value is 0x3B9A_C9FF. (RO)

Register 52.54. EMAC2NDUPT_REG (0x0710)

31																															0	
0x00000000																																Reset

TSSSUPD Configures the time to be initialized or added to the system time.

Measurement unit: second. (R/W)

Register 52.55. EMACNSUPT_REG (0x0714)

31	30	0
0	0x00000000	

Reset

ADDSUB with Configures whether to subtract or add the system time value the contents of the update register.

0: Add

1: Subtract

(R/W)

TSSSUPD Configures the sub-second representation of time, with an accuracy of 0.46 ns.

When EMACTSTPCTRL_REG[9] (TSCTRLSSR) is set, each bit represents 1 ns and the programmed value should not exceed 0x3B9A_C9FF. (R/W)

Register 52.56. EMACTSTPADDEND_REG (0x0718)

31	0
0x00000000	
Reset	

TSAR Configures the 32-bit time value to be added to the Accumulator register to achieve time synchronization. (R/W)

Register 52.57. EACTGTIME2ND_REG (0x071C)

31	0
0x00000000	
Reset	

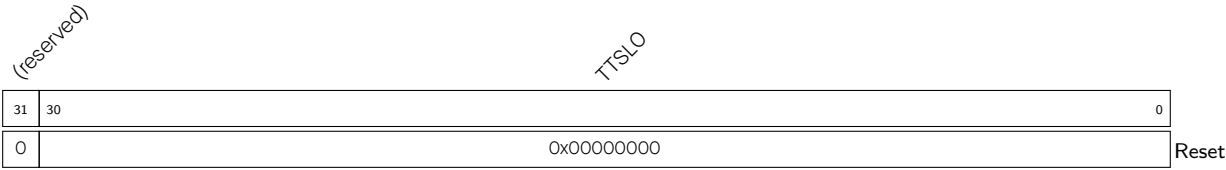
TSTR Configures the target time.

Measurement unit: second.

When the system time matches or exceeds the target time, an interrupt is generated (if enabled).

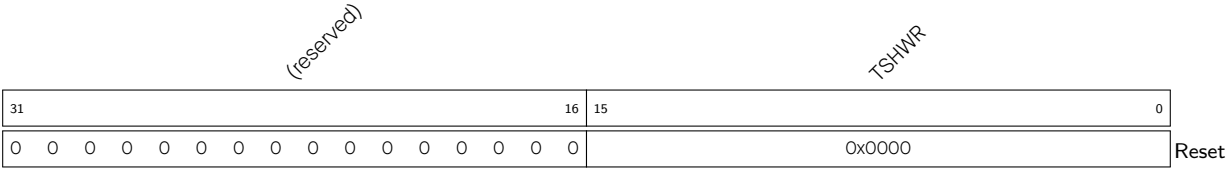
(R/W)

Register 52.58. EMACTGTIMENS_REG (0x0720)



TTSLO Configures the signed target time.
Measurement unit: nanosecond.
When the system time matches or exceeds the target time, an interrupt is generated (if enabled).
(R/W)

Register 52.59. EMACSYSTIMHIGHWORD2ND_REG (0x0724)



TSHWR Configures the most significant 16-bits of the timestamp seconds value.
The register is directly written to initialize the value.
This register is incremented when there is an overflow from the 32 bits of the System Time Seconds register. (R/W/SU)

Register 52.60. EMACTSTPSTATUS_REG (0x0728)

(reserved)																															TSTRGERR (reserved) TSTARGT TSSOVF																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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TSTRGTERR Represents whether the target time, being programmed in EMACTGTIME2ND_REG and EMACTGTIMENS_REG, is already elapsed.

0: Not elapsed

1: Elapsed

(R/SS/RC)

TSTARGET Represents whether the system time is greater or equal to the value specified in EMACTGTIME2ND_REG and EMACTGTIMENS_REG.

0: Less than the value

1: Greater or equal to the value

(R/SS/RC)

TSSOVF Represents whether the second value of the timestamp has overflowed beyond 0xFFFF_FFFF.

0: No overflow

1: Overflow

(R/SS/RC)